

GKBytes



Name

Class Section



7

Preety Rajpoot
(M.Tech)

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Preface



GK Bytes is a series of general knowledge books that are fountain head of knowledge for the children of class 1 to 8.

This series is as much about asking the right questions as it is finding out the right answer. It is about going beyond a few select subjects, and becoming more aware of the world around us, nothing things that are new, and changing, sharpening and feeding our curiosity, and learning to get to the bottom of things. Only a questioning mind can become a truly knowledgeable one.

Some of the features of this series are :

- New contents at each level are the extension of the class curriculum, with varying degrees of challenge, interest and colour.:
- Smooth and gently progressive gradation of contents.:
- New activities to encourage individual and group tasks.:
- Enhance language skills.

All in all, the new edition ensures that learning comes about at the initiative of the learner. It does this in a way that interactively and excitingly imparts information and knowledge which is closely related to the interests, the experience, the life and world of the child.

— Author and Publishers



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**CHAPTER**

Plants Adaptation

Plants have adaptations to help them survive (live and grow) in different areas. These adaptations might make it difficult for the plant to survive in a different place.

Read the descriptions and name the plants. You can pick your answers from the help box on the next page.

1. This plant gets its name from the fact that it literally comes back from the brink of drying up. It can survive long periods of heavy drought by curling up appearing dead. As soon as it gets a little water, it uncurls and reopens.



2. This tree has a life span of around 200 years and is a part of the lily family. It has a root system which consists of two roots. The deeper root system can have bulbs which store water up to 30 feet under the soil.

3. This tree survives in the harsh drier regions of south-western United States and Mexico because it has a huge swollen trunk base that stores a lot of water and thin and long leaves with a small leaf surface that help to control the loss of moisture.



4. This parasitic plant lacks true roots but have root-like structures buried in the branches of a tree. It depends upon the tree to anchor itself and absorb water and nutrients from its host. It has bright, large bird-pollinated flowers. Birds also disperse its seeds by eating its berries and depositing the seeds onto the branches of other trees.



5. These huge trees have green, glossy, and elliptical leaves. Older trees have huge aerial roots that can dangle down from great heights.

6. Orchids, bromeliads, and ferns fall in this category of plants that live on the surface of other plants, especially on the trunk and branches of the hosts. They grow on trees to take advantage of the sunlight.



7. These trees grow in the tropical deltas and along ocean edges and river estuaries. They have wide-spreading stilt roots to stay erect in the tidal mud and trap nutritious organic matter.

8. These plants are around 10-15 cm tall and have one flower per stem. These flowers follow the sun and the cup shape of the flower helps trap the sunlight. They are found in the Arctic regions.



Helpbox



Joshua Tree, Epiphytes, Elephant Foot Tree, Mangroves, Banyan Trees, Resurrection Fern, Arctic Poppy, Mistletoe



The monkey orchid has a strange adaptation which makes the petals of the flowers and the patterns resemble a monkey.



Activity

Collect some more interesting facts on plant adaptations.



Geographical Wonders of the World

The world is a masterpiece of God which is packed with astonishing and inspiring wonders. Once you come across with these natural marvels, their natural beauty will take your breath away.

In this grid, find the answers to the clues given below. Write them down in the blanks.

T	Y	B	O	R	A	B	O	R	A	Q	N	T	P
H	S	T	A	L	A	G	M	I	T	E	R	I	I
G	R	A	N	D	C	A	N	Y	O	N	S	T	N
A	U	C	G	A	L	A	P	A	G	O	S	I	E
T	S	T	A	L	A	C	T	I	T	E	W	C	A
A	P	G	A	N	G	O	T	R	I	D	Q	A	E
C	J	B	O	R	E	A	L	I	S	S	N	C	A
A	O	L	D	F	A	I	T	H	F	U	L	A	P
M	A	R	R	C	O	T	O	P	A	X	I	A	P
A	I	B	L	V	U	R	G	B	K	L	T	X	L
G	I	A	N	T	S	C	A	U	S	E	W	A	Y



1. A glacier located in Uttarkashi district, Uttarakhand, India which is the source of the River Ganga.
2. The most famous natural geyser located in Yellow Stone National Park, in Wyoming, USA.

3. A spectacular canyon carved by the Colorado River in Northern Arizona, USA.

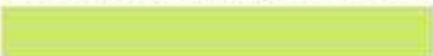
4. A cave formation or deposit on the floor of limestone caves, formed as the water drips from the ceiling of the cave and leaves tiny deposits of dissolved limestone.

5. One of the world's highest continuously active volcanos found in the Andes Mountains of South America.

6. This desert in South America is considered to be one of the driest deserts in the world.

7. These cave formations are limestone deposits that hang from the coolings of limestone caves.

8. Located at approximately 3,800 m above sea level on the border of Bolivia and Peru, this is considered to be the world's highest commercially navigable lake.

9. A volcanic island, part of the Society Islands of French Polynesia in the Pacific Ocean, surrounded by a lagoon and a barrier reef. At its centre are located the twin-peaks, Mount Pahia and Mount Otemanu.

10. This cluster of column cliffs, created by successive flows of lava, is located in the northern coast of northern Ireland. It is a great natural wonder of the United Kingdom.




Activity

Find out more geographical wonders and discuss them in the class.



CHAPTER

Ocean Kingdom

Two-thirds of our planet is covered by oceans. These oceans and their coastlines are amazingly rich in marine life. Can you identify these marine species?

1. A ten-legged crustacean, it can walk sideways and lives in the oceans as well as on the land. Its exoskeleton protects it from predators, besides providing support.



2. This deep-sea dweller has many light-producing organs along its body, especially in its belly. Its bio-luminescent organs light up as it swims through dark surroundings, helping it to attract and trap prey.

3. It has a streamlined body, full around the middle and narrowing to a paddle-shaped tail. With its wrinkled, grey-brown, sponge skin, it resembles the walrus.



4. This strong survivor is believed to have surfed through our oceans even before the dinosaurs walked on the Earth. Its skeleton is made of cartilage, which is a tough, fibrous substance. Its streamlined body shape helps it to glide smoothly through the water. It contains five to seven gill slits, which it uses for respiration.

5. Its body is composed of a bell-shaped jelly-like substance that encloses its internal structure. Its tentacles are covered with stinging cells that sting or kill other animals. The sting is often used to attack and catch prey, or as a defence mechanism.



6. This is the largest fish in the sea. Its favourite meal is plankton. This creature has a filter that sticks out its jaws and passively filters everything in its path.



7. This one ranges in colour from brown to orange to pink. Its body consists of a disk with a mouth in the centre of the underside, and a number of radiation arms or rays. Most of them can grow another arm if they lose one.

8. Also known as swellfish, blowfish, fugu, and globefish, it is characterized by its ability to puff up to about twice its normal size by gulping water. Different parts of its body such as liver, muscles, skin and ovaries contain a poisonous substance.



9. Its two hard shells and a soft body make it a bivalve. It stays mostly on the sea floor, squirting water out of its shells by rapidly opening and closing them.

10. They are the largest mollusc on Earth. They can grow to reach about 1.2 metres in length and can weigh more than 226 kg. They live in the warm waters of the South Pacific and Indian Oceans.



Activity

Find out more about enchanting marine habitat and understand the cause of its destruction.



CHAPTER

Study of Forest



Read the clues given about different kinds of forests and choose the correct answer from the help box.

1. This forest in California provides 60% of the state's drinking water and is an important carbon sink for the entire planet. Unfortunately it is threatened by pests, fire, urbanization, and mining interests.
2. This famous rainforest in Thailand is said to be over 160 million years old.
3. The legendary home of Robin Hood, this forest in England is famous for its wildlife and nearly 1,000 ancient oak trees.
4. These forests are found in cool rainy areas. Here the trees lose their leaves in fall or autumn and grow new leaves in spring.
5. The scientific study of forest species and their interaction with the environment is referred to as what?
6. These forests are dominated by pine and spruce trees. These forests have prolonged winters and get rainfall between 10 and 20 inches.

Helpbox



Sherwood forest, Deciduous forest, Sierra Nevada, Boreal forest, Forest ecology, Khao Sok



The floor of a coniferous forest is usually covered with a thick layer of dead leaves because they take a long time to rot in the cold, dark conditions.



Activity

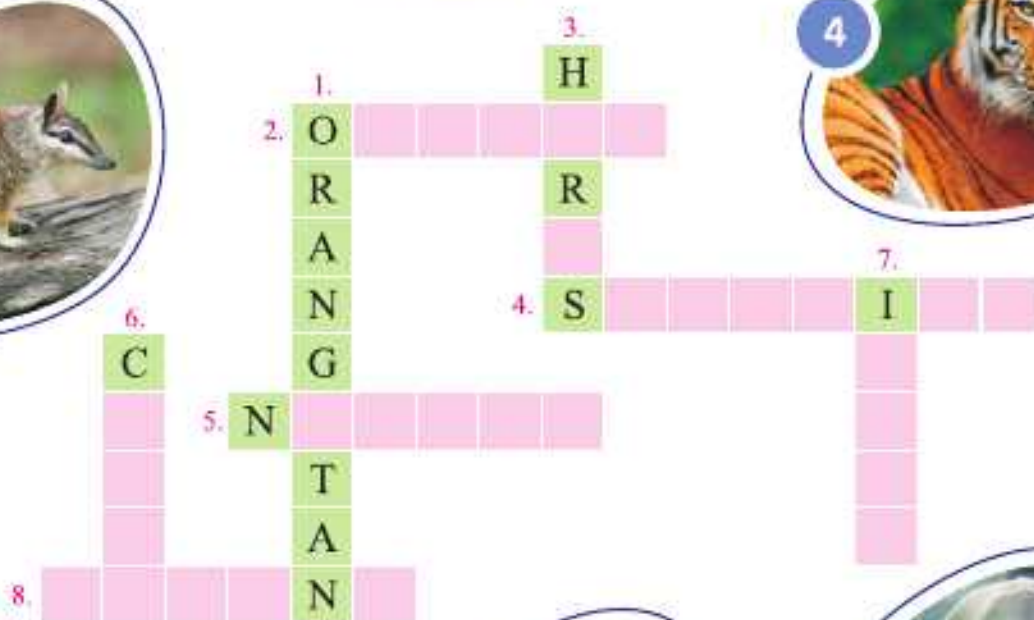
Find out, why are forests called the lungs of planet?

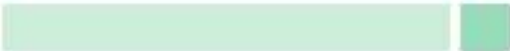
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CHAPTER

At A Great Risk

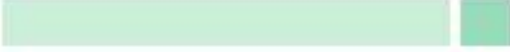
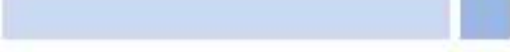
Complete the cross box with the help of the clues given on the next page.

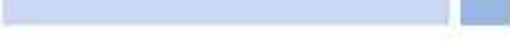


1. This endangered species is the only great ape living in the tropical rainforests of Borneo and Sumatra.

2. This medium-sized cat was once very valuable for its fur but is now protected by law.

3. Rhinos are hunted for this.

4. This species of tiger is the fewest in the world.

5. This now endangered pouchless marsupial lives in the open woodlands of western Australia.

6. This animal is found in Tibet and is hunted for its fleece which is known as Shatoosh.

7. Elephants are poached as their tusks contain.

8. Referred to as the 'sea cow' this endangered mammal lives in the sea.




According to a study by the International Union for the Conservation of Nature (IUCN) in 2008, one in four mammals is at a risk of disappearing forever.



Activity

1. Find out about other animals that are being hunted for the manufacture of luxury goods.
2. Prepare a report and make a presentation in the class.



CHAPTER

Living in Harmony

Explore nature, and you will find that all living things tend to share close bonds. They are interdependent and happily so! Meet some organisms that enjoy a symbiotic relationship – they work together to benefit from each other.

The **spider crab** lives in shallow areas of the ocean floor. A **greenish-brown algae** lives on the crab's back, making it blend with the environment, thus keeping it safe from its predators. While the algae gets a comfortable place to live, the crab gets camouflaged.



The **goby** used the shrimp's burrow for protection from predators. It also acts as 'eyes' for the pistol shrimp. The goby hovers above the burrow, feeding and interacting with other gobies. Meanwhile, the shrimp uses its antennae to stay in constant contact with the goby's tail while searching for food. If a predatory fish approaches, the goby flicks its tail several times, alerting the shrimp to retreat into the burrow. At night, the two rest together in the burrow.

The **tiny parasites** which are found on many ocean fish are very dangerous for their host fish and have the potential to cause many fish mortalities. While these parasites are very dangerous to the reef ecosystem, **cleaner wrasses** rely on them as their stable primary source of nutrients and food.



Oxpeckers are birds that ride on rhinos or zebras and eat ticks and other parasites that live on the host's skin. The oxpeckers get food and the host gets pest control. Also, when there is danger from approaching predators, the oxpeckers fly upward and hiss to alert their hosts of possible danger.

Aphids are small, soft-bodied, and defenceless insects that feed on plant sap. Ants find a colony of aphids and milk the waste plant sap from the body of the insects. In return, they protect the aphids from predators and parasites. In some cases, ants tend aphid colonies almost like ranchers tend their cattle, not only protecting the aphids but also moving them from plant to plant.





Honey badgers use their large and powerful claws to open beehives, where they feed on the honey, but are careful not to harm the larvae. Honey badgers have a symbiotic relationship with the honey guide bird. This bird leads the badger to a honey source.

QUIZ TIME

Tick (✓) the correct option.

1. Certain bacteria living in the intestines of human help them to produce :

(a) Vitamin A



(b) Vitamin D



2. What is commensalism?

(a) One species benefits and the other is not affected at all.



(b) One species benefits and the other is harmed in the process.



3. What is the most common form of mutualism?

(a) Eating symbiosis



(b) Predator symbiosis



4. Why is symbiosis important?

(a) It allow ecosystem to overlap and become dependent on each other.



(b) It kills all predators.



Activity

How can we live in harmony with nature? Discuss in class in group.

7

CHAPTER

Animals Adaptation

An adaptation is a way an animal's body helps it survive, or live, in its environment.

Can you identify these animals? Fill in the boxes to name them.

1. This member of the dog family can change its colour. Its fur is white or grey-blue during the winter, and grey-brown or grey during the summer, helping it camouflage. Its long, bushy tail, also called a sweep, helps it to change direction quickly and keeps its feet and nose warm when it curls up to sleep.



A r i x



2. Found in the rivers of eastern and central Africa, this animal is regarded as one of the most ferocious animals in the continent. It secretes a substance from its skin, popularly known as blood sweat, which acts as a natural sunscreen and protects it from sunburn. This secretion, however, is neither blood nor sweat. It prevents the growth of disease-causing bacteria.

i o a s

3. They are one of the six biggest kinds of snakes in the world. They can grow up to 4-6 metres in length. The largest ones are always females. Their hinged jaws can separate which allows them to swallow larger preys almost four to five times larger than their head.



B r m s y t n



4. This lizard has a skin flap which looks like a frill encircling its neck, which it can raise suddenly when threatened to surprise its enemy. This can make its head look several times its normal size.

r l L r d

5. Found in the tropical rainforests of South and Central America, this sluggish but curious mammal can be quite agile at times! Its long forelimbs which curved claws help it to cling to trees. Some species of this animal have hair that lets a greenish algae grow on them. The algae provides them with camouflage and essential nutrients.



S t



6. These frogs range from less than one inch to around two and a half inches in body length. They all vary in their colours and patterns. The beautiful colouring on their bodies are warnings to predators that they are poisonous. In certain cases, their patterns also help them to camouflage in their surroundings.

P s n D F r

7. This animal with black and white stripes on its face is the fastest burrowing mammal. During the cold winter months, its body temperature does not become as low as that of some other animals. It spends most of the day sleeping, and at night actively digs out other hibernating animals and devours them.



B g



8. This marine mammal lives in water and on ice in Antarctica. It can use its strong canine teeth to dig breathing holes in the cracks in the ice. It can also dive underwater and remain there for over an hour, hunting for fish, squid, and other marine animals. It can see well in dim light and uses its whiskers as a sense organ of touch.

W d l S a



Activity

Collect information about adaptation of birds and discuss it in the class.



CHAPTER

Adventures of Antarctica

Antarctica, on average, is the coldest, driest, and windiest continent, and has the highest average elevation of all the continents.

A. Try this quiz on the coldest, windiest, highest, driest and the loneliest of the continents.

1. Antarctica is in the _____ Hemisphere.
2. The biggest glacier in the world is the _____ in Antarctica.
3. The first explorer to reach the South Pole was _____.
4. The _____ in the Ellsworth Mountains is the highest mountain in Antarctica.
5. Rocky peaks of a ridge or a mountain exposed through an ice field or glacier are called _____.



B. Tick (✓) the sentences that are true, and cross (X) the ones which are false.

1. Seals are found in both, the Arctic and the Antarctic.
2. The polar bear is the largest animal in Antarctica.
3. Temperatures in Antarctica are generally colder than in the Arctic.
4. The native land fauna in Antarctica is completely invertebrate.
5. A 'whiteout' is an Antarctic weather condition, in which heavily falling or blowing snow results in poor visibility.



C. Choose the correct answers to these questions.

1. Gentoo, Adelie and Emperor are kinds of :
 (a) Sea lions (b) Seals
 (c) Penguins
2. Who navigated the first flight over the South Pole?
 (a) CB Eielson (b) Richard E Byrd (c) Lincoln Ellsworth
3. Killer whale is actually a kind of :
 (a) Shark (b) Dolphin (c) Whale



4. The commonest land animal in Antarctica is the :

(a) Mite



(b) Polar bear



(c) Penguins



5. The native people of Antarctica are called :

(a) Eskimos



(b) Inuit



(c) Antarctica has no native people



D. Though the living conditions in Antarctica are severe, some tough plants and animals live in and around the cold continent. Look at the pictures below and name these with the help of words given in the help box.

1.



2.



3.



4.



5.



6.



Helpbox



South Polar Skua, Emperor Penguin, Sea Lion, Krill, Baleen Whale, Fur Seal



Activity

Watch the movie The March of the Penguins to see a year in the life of emperor penguins.



CHAPTER

Deadly Disasters

Here is a quiz to test your knowledge on natural disasters.

1. Dry slab is the most dangerous form of v l c e .
2. u r n is not caused by shifts in tectonic plates.
3. Hypocentre is associated with an a h k .
4. The strongest earthquake in terms of magnitude after 1900s on record occurred in C i e .
5. A n o can produce the fastest winds on Earth.
6. The centre of a hurricane is called the e .
7. S fi i p n Hurricane Scale ranges from category 1 to 5. It uses the hurricane's speed to gauge size, damage, and flooding expected in an area.
8. K i River is known as the Sorrow of Bihar, as it causes widespread human suffering due to flooding every year.
9. In 2011, o h k earthquake was recorded as the most powerful known earthquake ever to have hit Japan.
10. On 28 August, 2005, a n a hurricane hit the Southern coast of the United States with devastating effect, killing more than 1,800 people.



Activity

Find out whether your school have a disaster management team or not. If yes gather all the information about the team.

10

CHAPTER

Environ Threats

To save, the environment, we need to be aware and more concerned.

Acid Rain is caused when industrial chemicals combine with condensing water vapour. The gases that cause acid rain usually come from automobiles or coal-burning power plants. It results in the acidification of lakes and streams, affects trees at high elevations, and damages sensitive forest soils. It also cause damage to monuments, statues and sculptures.



Water Pollution is one of the biggest sources of pollution, the planet is facing in the recent times. Besides garbage that makes its way into bodies of water, water pollution comes from many sources, including wastewater from industries, improper disposal of chemicals, sewage from cities that is emptied out into the water, and even sedimentation. This leads to the depletion of marine life.

Everyday, huge amounts of **hazardous wastes** are released or dumped into the environment. Old Batteries and medicines, shoe polish, paint things, etc. are some household hazardous waste. Paper, plastic, chemical, metal and rubber goods industries are sources of industrial hazardous waste. Hospitals generate a substantial amount of toxic waste too.



The warming of the Earth's surface and the lowest layer of the atmosphere due to the presence of water vapour, carbon dioxide, methane, and some other gases is known as the **greenhouse effect**. This is an essential process that allows life on Earth. But some scientists believe that increased and unregulated human activities may lead to the presence of excessive and harmful levels of greenhouse gases in the atmosphere.

QUIZ TIME

Answer the following questions.

1. Name the type of chemical that many deodorants release which is harmful for the ozone layer.
2. Over which continent was a massive hole first discovered in the ozone layer in 1985?
3. Which of the following chemicals led to the Bhopal Gas Tragedy in 1984?
(a) Carbon monoxide ☐ (b) Methyl Isocyanate ☐
(c) Chlorine ☐
4. Warmer climate results in the rise of sea level due to :
(a) Melting of glaciers ☐ (b) Expansion of oceans ☐
(c) Both (a) and (b) ☐
5. Which climate change agreement was signed in 2010 by a large number of countries under the UNFCCC?
6. Name the form of agriculture that is commonly practiced in the hills.
7. Which one of these diseases can arise due to stagnant water?
(a) Malaria ☐ (b) Tuberculosis ☐
(c) Influenza ☐
8. Which among the following is the most energy-efficient option?
(a) Incandescent bulb ☐ (b) Fluorescent lamp ☐
(c) Compact fluorescent lamp ☐
9. Name the main chemical constituent of biogas.



Project Work

Prepare a presentation on 'Pollution'.

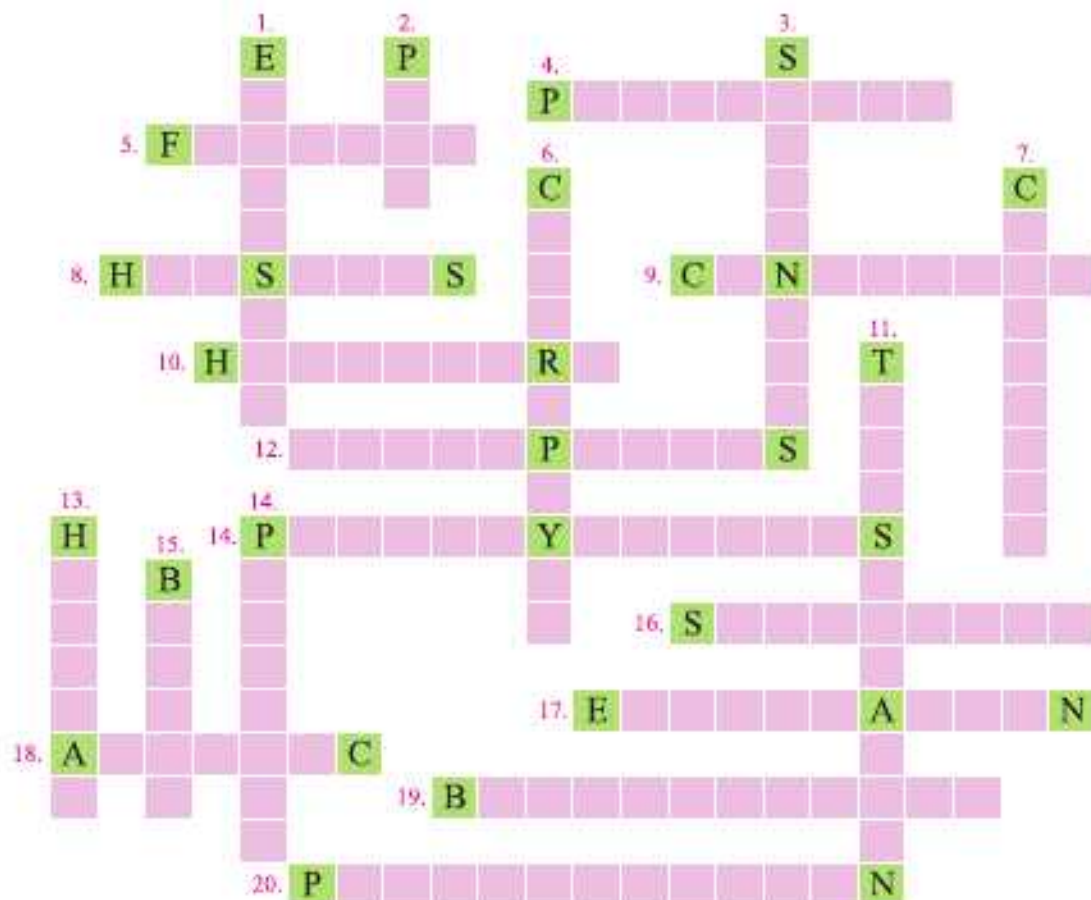


CHAPTER

Ecosystem Crossword

An ecosystem is a community of living organisms in conjunction with the non-living components of their environment.

Solve the crossword puzzle with the help of hints given.



Across :

4. Green plants that are the only members of a food chain that can produce their own food.
5. Network of related food chains.
8. Ecosystems where species are endangered.
9. Organisms that got nutrients by eating plants or animals, or both.
10. A type of consumer that eats plant material.
12. Tiny plants and animals that break down dead materials into nutrients.

14. The process plants use to make food using sunlight, water and carbon dioxide.
16. A mutual relationship between organisms that helps them survive.
17. Change of water from liquid to gaseous state.
18. Non-living factors that affect organism survival, such as availability of water and temperature range.
19. The number of different organisms in an ecosystem or a specified geographic region.
20. A process in the water cycle through which water vapour from clouds returns to the Earth.

Down :

1. A complex set of relationships among plants, trees, animals and people in the area.
2. An animal killed for food.
3. Consumers that feed upon organisms that other organisms have killed.
6. Green pigment in plant leaves.
7. A flesh-eating consumer.
11. Water lost to the atmosphere by the activities of plants.
13. Natural home of a plant or animal.
14. An animal that kills other animals for food.
15. Living components that directly or indirectly affect an organism in its environment such as disease, predators and parasites.



Activity

Find out how a soil ecosystem look like.



Corona Virus

Tick (✓) the correct answer.

1. What is the name given to the disease caused by corona virus spread in when?
(i) Covid - 19 ☐ (ii) China virus ☐
(iii) Corona - 19 ☐ (iv) None of these ☐
2. What is the name given by scientists to coronavirus before Wuhan?
(i) China virus ☐ (ii) Sars - cov - 2 ☐
(iii) Covid - 19 ☐ (iv) None of these ☐
3. Who has declared Covid - 19 an epidemic?
(i) WHO ☐ (ii) UN ☐ (iii) China ☐ (iv) India ☐
4. In which state has full curfew been imposed due to coronavirus?
(i) Punjab ☐ (ii) Haryana ☐ (iii) Rajasthan ☐ (iv) Kerela ☐
5. On which day is the public curfew "Janta Curfew" Imposed due to Coronavirus?
(i) 22 March 2020 ☐ (ii) 23 March 2020 ☐
(iii) 24 Mach 2020 ☐ (iv) 20 March 2020 ☐
6. Which state became the first state to lockdown due to coronavirus?
(i) Rajasthan ☐ (ii) Punjab ☐ (iii) Uttrakhand ☐ (iv) Gujrat ☐
7. Covid - 19 affects which part of the human body?
(i) Heart ☐ (ii) Lungs ☐ (iii) Brain ☐ (iv) None of these ☐
8. What is the name of the test of coronavirus (Covid - 19)?
(i) ELISA ☐ (ii) RT - PCR ☐ (iii) GFR Test ☐ (iv) None of these ☐
9. Coronavirus has started from Wuhan in China, it is located in which province of china?
(i) Hubei ☐ (ii) Shangai ☐ (iii) Shenzen ☐ (iv) None of these ☐
10. What does Vi in the world Covid - 19 mean?
(i) Corona ☐ (ii) Virus ☐ (iii) Diseases ☐ (iv) 2019 ☐



CHAPTER

Our Body

You see your body every day. It does so many actions : eating, running, walking, crying, shouting, writing, fishing, carrying, swimming, jumping and much more. Our body works like a machine. It works when we are awaking and when we are sleeping as well.

Each part of the body is made up of tiny structures called cells. The size of a cell is so small that a bunch of wells can be fit into a letter O. Many cells together form a tissue. Tissue of the some bind join together to form an organ. Organs that perform particular job makes a system. Different systems work together and keep our body alive.

Let us learn the major functions of some organs and systems of our body.

Sense Organs

Our body have five sense organs: eyes, ears, nose tongue and skin.



- ❖ Our eyes help us to see.
- ❖ Our ears help us to hear.
- ❖ Our nose help us to smell.
- ❖ Our tongue help us to taste.
- ❖ Our skin helps us to feels.
- ❖ Our sense organs make us aware of our surroundings. They are connected to the brain through the nerves.

14

CHAPTER

Science Scholars

Read the clues and guess the names of the famous scientists given below.

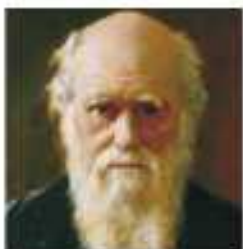
1. The Indian scientist won a Nobel Prize for Physics in 1930. He discovered a phenomenon that have been named after him.



2. He invented the analytical engine, which is the precursor to the modern electronic computer.



3. This scientist went to school only till he was ten years old. He also liked playing pranks on people with kites and invented the lightning conductor.



4. His book 'Origin of Species' brought forward the theory of evolution.



5. The English mathematician and astronomer elaborated upon the theory of gravity and also discovered the colour spectrum and the mathematical method we all use-calculus.

Harriet Brooks was the first Canadian woman nuclear physicist. She contributed significantly to study of radioactivity.



Activity

Find out who is referred to as the 'Missile Man of India' and also find out about his discovery.



CHAPTER

Unique Universe

A. Choose the correct answers to these questions about our unique universe.

- _____ is one of the many satellites of Jupiter. It is the heaviest natural satellite of Solar System. Its mass is nearly double of that of the Moon.
(a) Europa ☐ (b) Callisto ☐ (c) Ganymede ☐
- Vostok, Soyuz, and Apollo 13 are all names of _____.
(a) Lunar missions ☐ (b) Solar missions ☐ (c) Space crafts ☐
- This planet can produce the most violent winds and suffers the most violent weather in the Solar System _____.
(a) Jupiter ☐ (b) Neptune ☐ (c) Uranus ☐
- This planet does not spin on its axis like other planets but rather spins in a barrel sort of formation.
(a) Saturn ☐ (b) Mercury ☐ (c) Uranus ☐
- Kalpna Chawla was the Mission Specialist sent by NASA aboard the ill-fated space shuttle _____, and the first woman of Indian origin in space.
(a) Atlantis ☐ (b) Endeavour ☐ (c) Columbia ☐
- The Milky Way consists of at least 200 billion and may be as many as 400 billion stars. It is spread out as a thin disc and appears to be a spiral galaxy. It is about 100,000 light years across. Our _____ is located midway in the disc.
(a) Sun ☐ (b) Earth ☐ (c) Moon ☐



B. Name the following.

1. The Hubble Telescope have never managed to observe this planet.
2. The planet that is called the Earth's twin.
3. This planet has the largest volcano in our Solar System.
4. This layer of a gas surrounds the Earth
30 km above its surface.
5. Alpha, Beta, Gamma, Delta and Epsilon
are rings encircling this planet.
6. What name was given to the first series
of satellites launched in 1957 by the
Soviet Union?
7. In the Solar System, what are classified into groups, stones, irons and stony irons?
8. The study of what makes up the universe and how it has grown and changed?
9. The spot on the moon where humans first landed.
10. What name, literally meaning 'snowstorm' was given to the first reusable spacecraft
launched by USSR in 1988?



Mercury Surface, Space Environment, Geochemistry, and Ranging (Messenger) is a NASA Spacecraft, involved in a scientific investigation of the Planet Mercury.



Atoms and Molecules

Atoms and molecules are tiny, yet they form the core essence of any substance. Sometimes, a tiny part of a large substance can define its core essence.

Write the correct letters in the boxes to match the columns.

- | | | |
|--------------------|---|--|
| 1. Atom |  | (a) Have a positive electrical charge |
| 2. Element |  | (b) The smallest piece of matter that has the characteristic properties of a chemical element |
| 3. Nucleus |  | (c) Simplest and lightest atom with one proton |
| 4. Protons |  | (d) The total number of protons and neutrons in an atomic nucleus |
| 5. Electrons |  | (e) Made up of only one type of atoms |
| 6. Hydrogen Atom |  | (f) Have a negative electrical charge |
| 7. Periodic Table |  | (g) Consists of protons and neutrons |
| 8. Mass Number |  | (h) Smallest units into which a compound can be divided without changing its chemical properties |
| 9. Molecule |  | (i) Arrangement of elements in order of their atomic numbers |
| 10. Polar Molecule |  | (j) A molecule having a pair of electric charges or magnetic poles of opposite sign or polarity, separated by a small distance |

17

CHAPTER It Queries

Find out the correct answers from the word grid and encircle them. Use the clues given below.



A	H	A	C	K	E	R	T	D	V	T	T	O	A	O
M	S	Q	C	J	I	O	D	T	B	S	G	U	E	W
O	D	H	A	C	T	K	E	G	G	I	V	H	Z	I
N	C	K	I	R	A	A	S	O	E	L	F	J	P	N
I	V	D	G	F	G	M	A	I	A	I	V	Y	O	D
T	B	V	N	A	T	O	U	U	C	C	B	T	U	O
O	U	N	B	D	D	I	A	U	O	N	G	Q	W	
R	I	V	I	O	U	E	V	Q	I	N	J	C	Z	S
A	O	B	N	M	K	M	A	E	O	I	U	R	F	T
E	G	Q	O	E	P	U	O	D	J	U	I	Z	E	U
D	U	N	I	V	A	C	V	X	K	T	I	B	M	J
C	H	U	I	J	K	O	L	D	E	R	T	E	W	A
B	M	E	G	A	B	Y	T	E	H	V	O	J	K	L



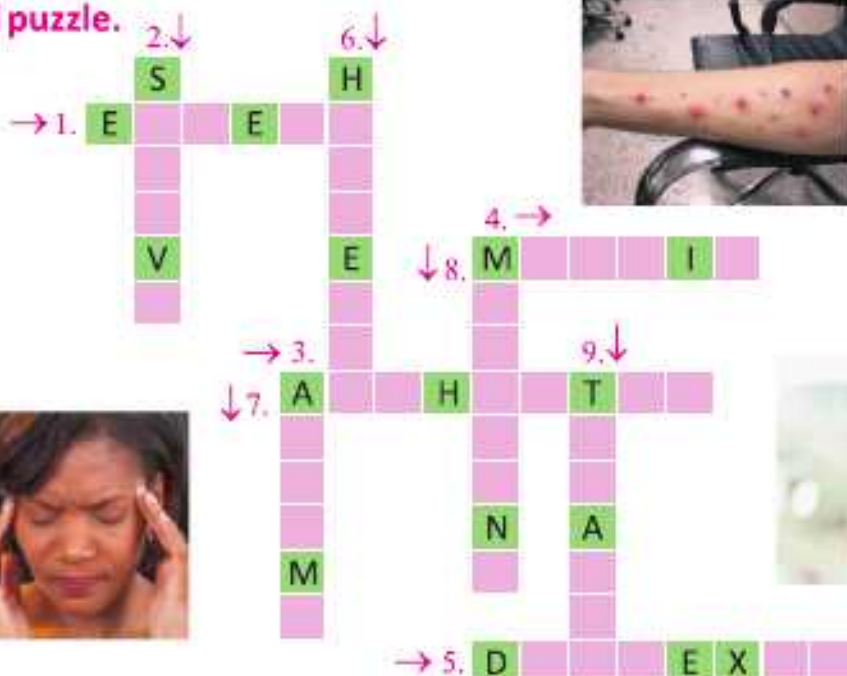
1. This operating system was introduced by 'Bill Gates' company, Microsoft. _____
2. Which element is the primary component in computer microchips? _____
3. On a computer keyboard, which key would you keep pressed if you have to type capital letters? _____
4. Which word describes a person who gains illegal access into computers? _____
5. What component of the computer comes in standard sizes 14, 15 and 17 inches? _____
6. The computer was the first to be mass produced. _____
7. In computer technology, what does MB stand for? _____
8. Big Blue is the nickname of which company? _____
9. Which device enables computers to communicate with each other across telephone cables? _____

18

CHAPTER

Health Problems

Solve the crossword puzzle.



Across: →

1. A kind of skin allergy in which the skin becomes itchy, red and dry.
3. This is what happens when joints get swollen, stiff, and painful due to the loss of flexible cushion between the bones.
4. A condition in which someone can see things that are up close, but finds it difficult to see objects that are far away.
5. Some children have this learning problem which can make words and ideas jumbled.

Down: ↓

2. The deficiency of Vitamin C leads to bleeding and swollen gums, weight loss, joint pains, and delay in the healing of wounds.
6. An allergy triggered by seasonal allergens, such as tree pollen.
7. A nebulizer is a machine that turns liquid medicine into a mist that can be breathed in. It is used by people who suffer from ailment.
8. Severe headache, usually on one side of the head, and often associated with nausea.
9. An acute, sometimes fatal, disease of the central nervous system caused by a bacterium which usually enters the body through an open wound.



Activity

Find out which countries have the highest rates of deficiency diseases and why?



CHAPTER

Branches of Medicine

What kind of doctor would you call to diagnose the following?

1. Johnny has a skin infection. He should consult a
2. Radha's grandmother who is aged needs a check up. Radha should take her to a
3. Amol's father has heart problems. He should consult a
4. Mohit is having a problem in digesting things. He should go to a
5. Gobind's mother's hormone levels are not balanced. She should consult an
6. Farhan is suspected to suffer from blood cancer. He should go to a
7. Savita's kidneys are not functioning too well. She should consult a
8. Karan's sister has breast cancer. She should be taken to an
9. Rohit's kids need to go for a check up. He should take them to a
10. Jagat is having breathing problems. He needs to get his lungs checked by a

Helpbox



Endocrinologist, Paediatrician, Oncologist, Haematologist, Cardiologist, Gastroenterologist, Dermatologist, Nephrologist, Pulmonologist, Geriatrician



Nursing is a well-respect medical profession. Florence Nightingale is known to have established nursing as a career and devoted her life to improving hospital care. She started the first nursing school.



Activity

Find out why Florence Nightingale was called the Lady with the Lamp!



Quiz on Chandrayaan-3

Chandrayaan-3 is indeed one of the most iconic missions in the history of Indian Space Research. Launched by ISRO, it is the 3rd and the most recent Indian lunar exploration mission.

Read and learn the following question and answer.

Q.1. When did the Chandrayaan-3 was launched?

Ans. 14th July 2023

Q.2. From where Chandrayaan-3 was launched?

Ans. Satish Dhawan Space Centre second Launch Pad in Shriharikota, Andhra Pradesh.

Q3. What is the name of the lander module of Chandrayaan-3?

Ans. Vikram.

Q4. What is the name of the rover module of Chandrayaan-3?

Ans. Pragyan.

Q5. When did the Vikram lander get separated from the propulsion module in Chandrayaan-3?

Ans. On 17th August 2023.

Q6. Which agency manufactured and launched Chandrayaan-3?

Ans. ISRO.

Q7. What is the full form of ISRO?

Ans. Indian Space Research Organisation.

Q8. What is the SATCAT No. (Satellite Catalog Number) assigned to Chandrayaan-3?

Ans. 57320.

Q9. What is the name of the authority which assigned SATCAT No.?

Ans. United States Space Commander.

Q10. Who is the chairman of ISRO?

Ans. Somanth.

Q11. Who is the mission director of ISRO?

Ans. Mohana Kumar

Q12. When did Chandrayaan-3 enter lunar orbit?

Ans. 5th August 2023.

Q13. What is total payload mass (weight) of Chandrayaan-3?

Ans. 3900 Kg.

Q14. What is the payload mass (weight) of the lander module of Chandrayaan-3?

Ans. 1752 Kg.



21

CHAPTER G-20

The G20 or Group of 20 is an intergovernmental forum comprising 19 sovereign countries, the European Union (EU), and the African Union (AU). It works to address major issues related to the global economy, such as international financial stability, climate change mitigation and sustainable development.

India Hosts G20 Summit:

Delhi decked up in its finest as India played host to leaders from across the world during the G20 summit on September 9-10. Roads were spruced up, fountains installed and pavements revamped as the national capital underwent a make over ahead of the big meet. Among the key outcomes of the summit was the inclusion of African Union into the grouping. The global Biofuel Alliance, a move towards adopting sustainable biofuels, were also launched. The big takeaway for India was the agreement on an economic corridor linking India, Gulf countries and Europe.

Who is in the G20?

The G20 is a forum comprising nineteen countries with some of the world's largest economies, as well as the European Union (EU) and, as of 2023, then African Union (AU). The countries are: Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, Mexico, Russia, Saudi Arabia, South Africa, South Korea, Turkey, the United Kingdom (UK), and the United States. Spain is invited as a permanent guest.



22

CHAPTER

Famous Indian Personalities

Look at the given pictures and write their name with the help of help box.

1



2



3



4



5



6



7



8



9



Helpbox



A, A, Dr. Meghnad Shah, Vikram Sarabhai, Salim Ali, Prof U.R. Rao, Dr. A.P.J Abdul Kalam, Dr. Jagdish C. Basu, Dr. J.J. Chinaty, Prof. K.R. Ramanathan, Dr. C.V. Raman, A



Lok Sabha Election Procedure

The Lok Sabha elections, also known as the general elections in India, are conducted to elect members of Parliament (MPs) who represent the people in the Lok Sabha, the Lower House of Parliament.

Election Schedule Announcement:

The Election Commission of India (ECI) announces the schedule for the Lok Sabha elections. The schedule includes the dates for various phases of Rolling. Counting of votes, and declaration of results.

Voter Registration:

Eligible citizens are required to register themselves as voters. The ECI conducts various voter registration drives to ensure maximum participation.

Candidate Nomination:

Political parties and independent candidates nominate their candidates for the Lok Sabha seats. Nomination papers are submitted to the returning officer of the respective constituency within the specified time frame.

Scrutiny of Nominations:

The returning officer scrutinises the nomination papers to ensure that candidates fulfil all eligibility Criteria and there are no discrepancies. Objections, if any, raised against nominations are also addressed during this process.

Withdrawal of Candidature:

Candidates have the option to withdraw their nomination papers within a specified period after scrutiny. Once the withdrawal process is completed, the final list of candidates is prepared.

Polling:

The Lok Sabha elections are conducted through electronic voting machines or EVMs and voter - verified paper Audit Trail or VVPAT machines. Polling takes place across multiple phases in different constituencies as per the election schedule.

Counting of votes:

After the completion of all phases of polling, the votes cast are counted. The counting process is supervised by the ECI and conducted in the presence of candidates and their agents.

Declaration of Results:

Once the counting of votes is completed, the election commission announces the results for each Lok Sabha Constituency. The Candidate securing the highest number of valid votes is declared the winner and becomes the MP.

Formation of Government:

The political party or coalition that secures a majority of seats in the Lok Sabha forms the government. The leader of the majority party or coalition is invited by the President of India to become the Prime Minister.

Oath-taking:

Elected MPs take an oath of office before assuming their roles and responsibilities in the Lok Sabha.



24

CHAPTER



UNIT 4

Sports and Entertainment

Cricket Twenty-20

Twenty 20 cricket was originally introduced by England and Wales Cricket Board in 2003 for professional inter country competition. In a twenty 20 game the two teams have a single innings each, which is restricted to a maximum 20 overs.

Find out how much you know about the Twenty 20 cricket matches by choosing the correct answers.

- What is the maximum number of overs a bowler can bowl in Twenty 20 cricket?
(a) Five ☐ (b) Four ☐
- Who made the first World Cup hat-trick against Bangladesh?
(a) Brett Lee ☐ (b) Ricky Ponting ☐
- The 2014 Twenty 20 World Cup was won by :
(a) India ☐ (b) Sri Lanka ☐
- Pakistan won the 2009 Twenty 20 World Cup by 8 wickets. Who were the runners-up?
(a) Sri Lanka ☐ (b) India ☐
- Which team won the 2014 ICC Women's World Twenty 20?
(a) England ☐ (b) Australia ☐
- A Twenty 20 Cricket match is played with 20 overs in each inning. What is the duration of each inning?
(a) 75 minutes ☐ (b) 90 minutes ☐



The first Twenty 20 Cup in 2003 was marketed with the slogan 'I don't like cricket, I love it'. It was meant to be a shortened and exciting form of the game that would bring the entire family together for an evening of fun and entertainment.



Activity

Find out what's one thing common between Twenty 20 and soccer?

25

CHAPTER

True Champions

Any man can win when things go his way. It's the man who overcomes adversity that is the true champion.

Do you know these true champions? Pick your answers from the help box.

1. Paralyzed with polio when she was a child, she proved that one can overcome obstacles and succeed through courage and determination. Her 1960 Olympics performance of winning three gold medals remains a stand-out performance in the history of the Olympics.



2. In 2001, he became the youngest person ever to climb Mount Everest. He lost five fingers due to frostbite during his first attempt to climb the mountain. However, this did not hold him back from making yet another very brave attempt.

3. She was a national level volleyball player. After being pushed out of a running train by thieves in 2011, one of her legs had to be amputated below the knee. She is the first female amputee to climb Mount Everest.



4. A former Major League Baseball pitcher, he played the game despite being born without a right hand. He was honoured as the United States flag bearer at the Pan American Games.

Helpbox



Jim Abbott, Temba Tsheri Sherpa,
Arunima Sinha, Wilma Rudolph



Activity

Share stories with your classmates of people who inspire you.



CHAPTER

Sports Man

A sportsman is a man who engages in sports, especially in some open-air sport, as hunting, fishing, racing, etc.

Identify these sports people whose distinguished contributions have made them legends.



1. Flawless on the clay court, she won 2 Australian Open, 7 French Open, 3 Wimbledon, and 6 US Open Singles titles.
2. He won gold medals in all 7 swimming events in which he participated, at the 1972 Olympics in Munich, Germany.
3. The first Asian to win the Wimbledon Junior Singles title, he was the World No. 3 tennis player in 1960s.
4. He ranks among the top welter weights and middle weights of all time, and was inducted into the International Boxing Hall of Fame in 1990.
5. One of England's most successful opening batsmen, he played 108 test matches, scoring 8114 runs at an average of 47.72.

Helpbox



Geoffrey Boycott, Ramanathan Krishnan, Chris Everts Mills, Sugar Ray Robinson, Mark Spitz



Activity

Prepare a list of all Indian sportsman of all fields.

27

CHAPTER

Martial Arts

The word 'martial' is derived from Mars, the Roman God of War. 'Martial Art' refers to the various systems of training for combat sports.

Read each of the descriptions and name the martial art.

1. An Olympic event, its name means 'the gentle way'. This Japanese art form focuses primarily on throwing and ground-fighting techniques.



2. This is a Russian martial art and sport. The name itself is an acronym. It has roots in judo and styles of wrestling.

3. A standing sport that combines styles of boxing and kicking. It is often practised for self-defence and general fitness.



4. This Brazilian martial art combines martial arts and dance. Its bouts are accompanied by lively music and allow the fighter to show-off their skill.

5. The national sport of South Korea, it stands for 'art of kicking and punching'. It is known for its striking and athletic kicks.



Helpbox



Capoeira, Judo, Kick boxing, Tae Kwon do, Sambo



Activity

Do you know anyone who practises a traditional martial art form of your country? Talk to the person to find out the goals, techniques, and benefits of the art form.



Athletics

Athletics is an exclusive collection of sporting events that involve competitive, running, jumping, walking.

Choose the correct option.

- Which of these is not a throwing event?
(a) Discus ☐ (b) Javelin ☐ (c) Hurdling ☐
- Which athletic event is Vikas Gowda of India most well-known for?
(a) Discus throw ☐ (b) Shot-put ☐ (c) Triple jump ☐
- Which event was once known as hop, step, and jump?
(a) Pole Vault ☐ (b) Triple Jump ☐ (c) Relay Race ☐
- Which of the following athletic events was Sergey Bubka well-known for?
(a) Pole Vault ☐ (b) High Jump ☐ (c) Long Jump ☐
- Which of these is a team sport in athletics?
(a) Relay Race ☐ (b) Shot-put ☐ (c) Pole Vault ☐
- Which athletics event combines long-distance running with hurdling, each runner being required to clear seven water jumps and 28 hurdles in a 3,000 metre course?
(a) Marathon ☐ (b) Biathlon ☐ (c) Relay Race ☐
- In which athletic event does the athlete take a running jump to attain height?
(a) High Jump ☐ (b) Decathlon ☐ (c) Pole Vault ☐



Milkha Singh is regarded as one of the best athletes of India, and was nicknamed the 'Flying Sikh'. He won two gold medals in the 1958 Asian Games and a gold medal in the 1958 Cardiff Commonwealth.



Group Discussion

Find the names of some current Indian athletes and their achievements & discuss in group in your class.

Adventure Sports

- A. Adventure sports include everything from extremely dangerous, life-threatening sports to more common travelling activities. Name these adventure sports. Pick your answers from the help box.



1



2



3



4



5



6



7



8

Helpbox



Ice climbing, Bungee jumping, Sand boarding, Surfing, Paragliding, Skateboarding, Skiing, Kayaking

QUIZ TIME

B. Choose the correct option to name the sport being described in the clues.

1. You are wearing a mask under water. You are doing :
(a) Snorkeling ☐ (b) Sailing ☐
2. You are exploring the rough terrain in and around caves. You are doing :
(a) Spelunking ☐ (b) Ballooning ☐
3. Your skates have blades. You are doing :
(a) Skiing ☐ (b) Ice skating ☐
4. Ropes are extremely important to you. You are doing :
(a) Mountain Climbing ☐ (b) Rafting ☐
5. You are using a vehicle to do this activity. You are doing :
(a) Parasailing ☐ (b) Scuba diving ☐
6. You often go cross-country. You take part in :
(a) BMX biking ☐ (b) Jet skiing ☐



Motocross is a type of motor cycle racing. The motorcyclists race in closed courses laid out on rough terrains. This race has its origin in Great Britain. Motocross is known to be a difficult motorcycle race.



Activity

What is your favourite adventure sport? Does this count as an adventure sport anyway?



Celebrities Who Care

Do you know these celebrities? Choose your answers from those given in the help box.

1. This famous actor-director is regarded as a 'Messenger of Peace' for rousing people across the world to find out more about the crisis in Darfur, Sudan. 
2. A leading actress in Indian cinema. She is a supporter of the rights of slum dwellers. She have also campaigned against ostracizing the victims of AIDS. 
3. This Colombian singer, whose songs include 'Loca', has set up a foundation to help school children with their education in her homeland. 
4. He is one of the most successful entrepreneurs in the history of the world. He owns Microsoft and has set up NGOs all over Africa. 
5. This actor has starred in many martial arts films, such as 'Rush Hour' and 'Karate Kid', and has bequeathed half his assets to charity in his will which is estimated to be at \$64 million. 



Helpbox



Shakira, George Clooney, Jackie Chan, Bill Gates, Shabana Azmi



Muhammad Ali is a retired American boxer and three-time World Heavyweight Champion. Suffering from Parkinson disease, he has been involved in raising funds for the Muhammad Ali Parkinson Centre in Phoenix, Arizona. Over the years, he has also supported the special Olympics and the Make a Wish Foundation.

A. Match the painters with their paintings by writing the correct letter in the boxes.

- | | | |
|------------------------------|---|-----------------------|
| 1. Sunflowers |  | (a) Raja Ravi Verma |
| 2. Guernica |  | (b) Claude Monet |
| 3. Shakuntala |  | (c) Vincent Van Gogh |
| 4. Mona Lisa |  | (d) Salvador Dali |
| 5. Waterlilies at Giverny |  | (e) Pablo Picasso |
| 6. The Persistence of Memory |  | (f) Leonardo da Vinci |

B. Find the names of these Indian Painters in the word grid.

- Amrita Shergill
- Jamini Roy
- MF Hussain
- Nandalal Bose
- Anjolie Ela Menon
- Jatin Das
- Manjit Bawa



A	N	J	O	L	I	E	E	L	A	M	E	N	O	N
Q	A	N	J	J	A	A	C	N	W	R	M	C	R	J
J	M	S	N	Z	L	Q	Y	N	L	K	A	C	K	T
A	R	O	A	E	H	Q	C	I	A	M	N	M	B	R
T	I	S	N	F	I	D	I	Q	L	U	J	D	A	X
I	T	Z	D	J	D	Q	O	G	M	K	I	L	X	P
N	A	I	A	H	R	K	O	S	F	T	T	H	K	F
D	S	S	L	J	I	H	D	W	H	X	B	F	G	F
A	H	I	A	V	Y	E	L	H	U	D	A	I	O	C
S	E	W	L	W	I	X	V	T	S	N	W	D	X	H
K	R	I	B	M	B	Q	Q	Y	S	O	A	Q	E	S
D	G	G	O	T	N	Z	M	G	A	U	V	V	H	P
U	I	H	S	J	A	M	I	N	I	R	O	Y	D	F
H	L	G	E	A	H	H	S	J	N	M	A	S	M	S
A	L	P	Q	M	I	A	W	D	S	G	S	K	R	M



Activity

Find out more about the works and lives of these Indian Painters. Write a paper on each and read them out in the class. Display pictures of their paintings.

32

CHAPTER

Dance Form

Dancers are performing artists. Their varied colourful art forms have engaged and thrilled audiences since ages.



Puppetry seems to have existed in almost all civilizations since times unknown. Be it the Bunraku of Japan, or the marionettes of Italy, or the famous Rajasthani puppets of India, performances with colourful puppets are among the most favourite forms of storytelling.

Theatre involves live performances where a story is told theatrically to the audience watching. The ancient Greeks were the first theatre-goers. They gathered in hundreds and sat on hillsides to watch tragedies and comedies unfold before them. William Shakespeare is one of the most famous playwrights of the world.



Ballet was first developed as an art form in the 1600s in France. Grand orchestras, spectacular costumes, and stage scenery are essential elements of this graceful dance form. **Swan Lake** and **The Nutcracker** are some of the most famous ballets.

Odissi is one of the eight classical dance forms of India. It originates from the state of Odisha, in eastern India. It is a dance of devotion and replicates the graceful and sensuous postures found on classical temple sculptures. It is danced to expressive music such as **Geet Govind**.



Kathak originated as a court dance, and some believe it evolved from Lord Krishna's raas lilas.

Sattriya is one among eight principal classical Indian dance traditions. Whereas some of the other traditions have been revived in the recent past, Sattriya has remained a living tradition.



QUIZ TIME

Fill in the blanks.

1. A dummy is a puppet operated by a person. While the person says the dialogues, it appears as though the dummy is saying them. Name that person.



2. What is a full turn of the body on the point of the toe called in ballet?



3. The Poetics is the earliest surviving work of dramatic theory from ancient Greece. Who was its author?



4. An opera performance combines musical scores and a text. What is the text called?

5. In ballet, what is pas de deux?

6. Kuchipudi and Kathak are classical dances of which country?



7. Who wrote the famous play "A Midsummer Night's Dream"?



Activity

What are some of the famous native performing art forms of your country? Find it.



At the Movies

1. Which animated film is about an old man and his dream to live by Paradise falls?
2. Which was the first Indian 3-D movie?
3. In the movie 'Star Wars', what was Darth Vader's name before he becomes a cyborg?
4. Which river provides the backdrop in the movie 'Anaconda'?
5. In which movie does Julie Andrews play the role of Maria?
6. By what name was Norma Jean Mortenson better known?
7. Who directed the film "Bend it Like Beckham"?
8. Which Kamal Hassan film was inspired by the Hollywood film "Mrs. Doubtfree"?
9. Which actor played the role of Jai in the film "Sholay"?
10. Which country does Jackie Chan originally hail from?
11. Who plays Gandalf in the film series "The Lord of the Rings"?
12. In the movie "Mulan", what is the name of the dragon that helps Mulan to train for the war?



Khuda Kay Liye (2008) was the first Pakistani film to be officially released in India after about four decades. The film was directed by Shoaib Mansoor.





Centrestage

Choose the correct options.

1. Harry Houdini is one of the most recognisable names in which field of performing arts?
(a) Mime ☐ (b) Opera ☐ (c) Magic ☐
2. In which of these dance forms of India do dancers usually wear a mask?
(a) Kuchipudi ☐ (b) Manipuri ☐ (c) Chhau ☐
3. Who is believed to be the author and compiler of the ancient treatise of performing arts, 'Natya Shastra'?
(a) Manu ☐ (b) Bharata Muni ☐ (c) Vilhana ☐
4. In which of these parts of India is 'jatra' a popular form of theatre?
(a) West Bengal ☐ (b) Rajasthan ☐ (c) Kerala ☐
5. 'Noh' and 'Kabuki' are traditional theatrical forms of which country?
(a) China ☐ (b) Japan ☐ (c) Russia ☐
6. Who among these is not a well-known mime artist or silent actor?
(a) Jogesh Dutta ☐ (b) Charlie Chaplin ☐ (c) Criss Angel ☐
7. Which famous playwright, actor, and director directed the famous play "Charandas Chor"?
(a) Habib Tanvir ☐ (b) Arvind Gaur ☐ (c) Rajat Kapoor ☐
8. Who has written the famous plays "Dance like a Man" and "Final Solutions"?
(a) Mahesh Dattani ☐ (b) Girish Karnad ☐ (c) Arvind Gaur ☐

The Ancient Greeks were one of the first theatre-goers. They gathered in hundreds and sat on hillsides to watch tragedies and comedies unfold before them on stage.



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CHAPTER

Great Music Composers

Music is a combination of sounds that is usually pleasing to the ears. It can be produced by a singer, a musical instrument or even a wind chime. The Composer write musical notation, music theory, instrumentation and handling musical ensembles.

Choose from the help box and read the clues to name these famous music composers.

1. This famous sitar player and music composer played a key role in popularising Indian music in the western world. He has been awarded the Bharat Ratna and is the winner of 3 Grammy awards.
2. This Indian violinist, composer, and music conductor is internationally famous for his techniques of classical Carnatic as well as Western Classical Music compositions.
3. This famous musician from Assam sang about humanity and universal brotherhood. He was awarded the Padma Vibhushan posthumously in November, 2011.
4. This Grammy nominated film composer, jazz musician, and keyboardist is often known as the 'Godfather of Indian Jazz'.
5. He is perhaps one of the most well-known music composers of the Indian film Industry. His son Pancham, followed his legacy and became a popular music composer as well.



Helpbox



Sachin Dev Burman, Ravi Shanker, Bhupen Hazarika, Louis Banks, L Subramaniam



Activity

Do you know how to read music from musical notations? Or how to use these notations to create music? If you do, explain to your classmates how to do it.

**CHAPTER****FIFA World Cup in Qatar**

Tick (✓) the correct answer.

1. From when to when did the FIFA World cup 2022 take place in Qatar?
(i) From November 2 to December 18 ☐ (ii) From December 5 to December 18 ☐
(iii) From December 8 to December 28 ☐ (iv) From November 5 to December 18 ☐
2. Which country has hosted the most world cups?
(i) France ☐ (ii) Italy ☐ (iii) South Africa ☐ (iv) Germany ☐
3. Who won the best young player award in FIFA world cup 2022?
(i) Enzo Fernandez ☐ (ii) Lukas Podoiski ☐
(iii) Thomas Mulker ☐ (iv) Landon Donovan ☐
4. Who will host the 2026 world cup?
(i) Germany ☐ (ii) Australia and New Zealand ☐
(iii) France ☐ (iv) United states, Mexico and Canada ☐
5. Who invented Modern football?
(i) Pop warner ☐ (ii) John Heisman ☐
(iii) George Halas ☐ (iv) Walter Camp ☐
6. Which of the following team awarded FIFA Play Award?
(i) England ☐ (ii) France ☐ (iii) Brazil ☐ (iv) Argentina ☐
7. Nora Fatehi flaunts her dance moves in FIFA world cup 2022 anthem _____.
(i) Tatu Bom de Bola ☐ (ii) Live it UP ☐
(iii) We will find a way ☐ (iv) Light the sky ☐
8. Which of the following country finished third in the 2022 FIFA world cup?
(i) Morocco ☐ (ii) Argentina ☐
(iii) Croatia ☐ (iv) France ☐
9. What was the first world cup trophy called?
(i) Jules Rimet Trophy ☐ (ii) Stanley Rous Trophy ☐
(iii) Joao Havelange Trophy ☐ (iv) Stanley Rous Trophy ☐
10. In FIFA World cup 2022, Golden Ball Award goes to?
(i) Kylian Mbappe ☐ (ii) Lionel Messi ☐
(iii) Luka Modric ☐ (iv) Cristiano Ronaldo ☐



Business Basics

You must have heard the term 'Business'. What does it actually mean? Test your knowledge of the business world by picking the right answers to these questions.

1. What was the name of the person who founded facebook while attending college at Harvard University?
2. Which bank serves as the central bank of India?
3. If New York has Wall Street, what does Mumbai have?
4. Which global trading and investment company, founded in 1850, became bankrupt in 2008?
5. Who were the founders of the company Google?
6. A dairy cooperative in Gujarat, India, is jointly owned by about 2.6 million milk producers. What is the name of their brand?



Divyank Turakhia from India is a young entrepreneur who started his Internet consulting business at the age of 14 and launched his company at 16.



Project Work

Prepare a presentation on five Indian Businessmen.



Culture Curry

Culture refers to the pattern of human activity and the symbols, which give significance to this activity. The cultural values of a community give it an identity of its own.

Fill in the blanks with the answers. Choose the answers from the help box.

1. This country has many myths and legends pertaining to the actions of the Gods. One of the most famous families of Gods they worshipped were the children of Nut and Geb-Orisis, Isis, Seth and Nephthys. Which country are we talking about?



2. This dance form originates from Cuba and it is danced to music of the same name. It comes from studies made by dance teacher Monsieur Pierre Zurcher-Margolle. The authentic Cuban style can be very complex and involves polyrhythms.

3. Ayurveda, the earliest school of medicine known to mankind, has been an integral part of this country's tradition since ages. Mark Twain described the rich culture of this nation when he called it 'The Cradle of the human race, the birth place of human speech, the mother of history, the grandmother of legend, and the great grandmother of tradition'. Which country are we referring to?



4. This is a music festival that takes place every year near Somerset in England. In 2013, famous bands like The Arctic Monkeys and Mumford and Sons played here.

Helpbox



Oktoberfest, Paris, Joan of Arc, India, Glastonbury, Cha Cha Cha, Mali, Indonesia, Egypt, Ramlila

5. This city houses famous museums like the Louvre and Orsay Museums. It also houses the biggest collection of impressionist art in the entire world. Which city are we talking about?

--	--	--	--	--



6. This sixteen-day festival held annually in Munich, Germany, during the months of September and October is also one of the world's largest fairs. Name the festival.

--	--	--	--	--	--	--	--	--	--

7. This peasant girl led the French in the Hundred Years' War because she believed that it was God who guided her to do so. She was executed as a heretic and later made a saint. Who is she?

--	--	--	--

--	--

--	--	--



8. In northern India, this dramatic reenactment of the life of Lord Rama is staged for ten or more consecutive nights. On the last day of the performance, effigies of Ravana, his brother and son are set on fire, and the coronation of Rama takes place. What is this dramatic enactment called?

--	--	--	--	--	--

9. The Wayang Puppet Theatre has been practised in the royal courts of Java and Bali for centuries. It is an ancient form of storytelling. In which country is it practised?

--	--	--	--	--	--	--	--



10. Festival au Desert is three days of Touareg music, art and dance that are on display in Esskane, which is two hours away from Timbaktu. Artists from all over Africa and Europe come and camp in the desert. Which country hosts this festival?

--	--	--	--



Activity

Make a presentation on Asian countries culture.



Nobel Prize Winner

You must have heard the name of Nobel Prize. What does it actually mean? Nobel Prizes are given in six subject areas— Physics, Chemistry, Physiology or Medicine, Literature, Peace and Economics.

Match these Nobel Laureates with the fields in which they were awarded the prize by writing the correct letters in the boxes.

- | | | |
|------------------------|----------------------|----------------|
| 1. Henri Dunant | <input type="text"/> | (a) Peace |
| 2. Rabindranath Tagore | <input type="text"/> | (b) Peace |
| 3. Marie Curie | <input type="text"/> | (c) Physics |
| 4. Rudyard Kipling | <input type="text"/> | (d) Chemistry |
| 5. Max Planck | <input type="text"/> | (e) Literature |
| 6. Ernest Rutherford | <input type="text"/> | (f) Literature |
| 7. Barack Obama | <input type="text"/> | (g) Economics |
| 8. Amartya Sen | <input type="text"/> | (h) Medicine |
| 9. VS Naipul | <input type="text"/> | (i) Economics |
| 10. Har Gobind Khorana | <input type="text"/> | (j) Literature |
| 11. Paul Krugman | <input type="text"/> | (k) Medicine |
| 12. Francis Crick | <input type="text"/> | (l) Physics |

These prizes are named after Alfred Nobel, a Swedish scientist, who had invented the explosive, dynamite.



Activity

Make a list of all Indians who have won the Nobel Prize. Also write about each of their contribution. Talk about this in class.

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CHAPTER

Historic Monuments

A monument is a type of structure that was explicitly created to commemorate a person or important event, or which has become important to a social group as a part of their remembrance of historic times.

Identify these famous landmarks around the world. Pick your answers from the help box.

1. This National Martyr's Memorial in Bangladesh is dedicated to the courage and sacrifice to those killed in the Bangladesh Liberation War of 1971.



2. Following India's independence, this monument became the site of the Indian Army's Tomb of the Unknown Soldier.



3. This monument was built in St. Petersburg to commemorate the Russian victories over Sweden in the Great Northern War.



4. This memorial on one of the two rocks located in Kanyakumari is built in the honour of Swami Vivekananda.

5. These life-size clay soldiers, buried in the grave of the first Chinese Emperor Shihuangdi, were discovered in 1974 near Xi'an in China.



6. Located in Ottawa, Ontario, this memorial is dedicated to all the unidentified Canadian Soldiers and is part of the National War Memorial.



Helpbox



India Gate, Tomb of the Unknown Soldier, Monument to Peter I, Vivekanand Rock Memorial, Terracotta Army, Jatiyo Smriti Sudho



Activity

Read about 5 more monuments in India. Discuss in the class.

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CHAPTER

National Anthem

A National Anthem is a generally patriotic musical composition that evokes and eulogizes the history, traditions and struggles of its people, recognized either by a nation's government as the official national song or by convention through use by the people.



India and Bangladesh adopted two songs written by the Indian Nobel laureate and noted Bengali poet-author Rabindranath Tagore as their national anthems. Jana Gana Mana was adopted as the national anthem of India on 24 Jan, 1950. Amar Sonar Bangla became the national anthem of Bangladesh in 1971. Another Nobel Prize Winner who composed a national anthem was Bjørnstjerne Martinius Bjørnson. He wrote the lyrics for the Norwegian national anthem.



La Marseillaise is the French national anthem that was written by Claude-Joseph Rouget de Lisle, who was a captain of the engineers. It was composed in one night during the French Revolution in 1792.



QUIZ TIME

1. Though the Tamil version of the Sri Lankan national anthem is played on some occasions, it was originally composed in _____.
2. Mozart composed music for the national anthem of which country? _____
3. Which country has the shortest national anthem? _____
4. Who wrote the lyrics of The Star-Spangled Banner, the national anthem of the United States? _____
5. Which country's national anthem is God Save the Queen? _____

The study of anthems is called anthematology.



Activity

Collect some more information on national anthems.



Guess The Brand

A Brand is a name of trademark connected with a product to make consumers trust the product.

Here are some catchy brand taglines. Name the brands. Take help from the help box.

1. Between love and madness lies obsession
2. The world's favourite airline
3. The ultimate driving machine
4. We keep your promises
5. Expanding possibilities
6. Empowering the Internet Generation
7. Just do it
8. The world on time
9. Think different
10. The best a man can get
11. Like No Other
12. Connecting People
13. Enjoy the ride
14. Driven by Passion
15. Built for the road Ahead
16. Life's Good
17. Always low prices, Always
18. Sense and Simplicity





Activity

Find out more brand taglines along with their logo. Write them on a chart paper.

Helpbox



Philips, Walmart, LG, Ford, Fiat, Nissan, Nokia, Sony, Gillette, Fed Ex, Nike, Cisco, Hewlett Packard, DHL, BMW, British Airlines, Calvin Klein, Apple



CHAPTER

Grasslands

The most extensive natural grasslands are believed to be the intermediate regions, bordered by forests at one end and dry deserts at the other.

A. Match these grasslands with the places where they are found, by writing the correct letters.

- | | | |
|------------|---|----------------------------|
| 1. Llano |  | (a) The USA |
| 2. Pampas |  | (b) Arctic Circle |
| 3. Prairie |  | (c) Brazil |
| 4. Steppe |  | (d) Argentina |
| 5. Tundra |  | (e) Colombia and Venezuela |
| 6. Veld |  | (f) Central Asia |
| 7. Campos |  | (g) South Africa |



B. Choose the correct option.

- Grasslands can be _____.
 (a) Tropical ☐ (b) Temperate ☐ (c) Both (a) and (b) ☐
- Which of these crops are grown most extensively in the Prairies?
 (a) Rice ☐ (b) Wheat ☐ (c) Jute ☐
- What are the tropical grasslands called?
 (a) Prairie ☐ (b) Savanna ☐ (c) Taiga ☐
- Which of the following animals are not found in grasslands?
 (a) Tigers ☐ (b) Lions ☐ (c) Bisons ☐



Activity

In group, make presentations on the various forest of the world. Share your presentation with the class.



21st Century

Do a quick recap of some of the events and occurrences in the 21st century, by answering the following questions. Pick your answers from the help box.

1. Which Pakistani Sufi Rock band performed on the banks of the Dal Lake in Srinagar in 2008, to promote peace and brotherhood between India and Pakistan?
2. Which virus-inflicted epidemic was known to have started in Gunagdong Province, China, in November 2002 and later spread around the globe?
3. Amidst wide and violent protests from the Arab World, this country was formed and given legitimacy in 1946.
4. Which country launched Chandrayaan I into space on 22 October 2008, thus joining a select group of countries to send a spacecraft to the Moon?
5. The US attack in 2003 marked the end of Saddam Hussain's rule in this country.
6. Name the spaceshuttle that docked with the Hubble Space Telescope in 2009 for much needed repairs.
7. Which engineer and multimillionaire from the USA became the first space tourist in 2001, when he paid a reported \$ 20 million for a trip?



Helpbox



Junoon, Dennis Tito, India, Israel, Iraq, Atlantis, Severe Acute Respiratory Syndrome (SARS)



Activity

Find out 10 current affairs of your countries and discuss them in the class.



Language Word Search

E	A	R	T	Y	U	T	S	Y	V	J	I	O	M	H
A	P	B	V	X	V	I	E	I	M	A	G	E	R	Y
L	R	I	M	V	U	D	T	H	M	G	H	V	K	P
L	O	C	T	N	A	U	G	A	Q	I	A	I	G	E
I	L	V	S	A	T	I	R	E	U	O	L	E	W	R
T	O	G	P	I	P	Q	R	P	I	I	Q	E	Q	B
E	G	H	N	B	E	H	F	I	R	O	E	O	I	O
R	U	K	A	A	Q	A	S	T	T	M	A	S	C	L
A	E	R	S	W	E	I	U	H	G	B	V	D	F	E
T	O	W	F	E	P	A	O	E	C	A	E	U	H	O
I	A	Q	V	R	A	Q	Y	T	X	O	A	K	M	I
O	I	E	N	M	I	E	S	Y	N	T	A	X	O	P
N	U	A	L	L	E	G	O	R	Y	Q	V	B	I	L
A	Q	F	Y	H	U	A	S	D	R	T	U	I	A	J
V	U	V	T	N	J	O	X	Y	M	O	R	O	N	E

1. A poem or speech, usually at the beginning of a play, which introduces the contents
2. A figure of speech that combines contradictory terms
3. Use of the same sound or letter at the beginning of words that are close together in a phrase or sentence
4. The rules of grammar that describe the way words and phrases are put together to form sentences in a language
5. Language that creates pictures in the minds of people reading or listening
6. Use of exaggerated language to stress upon the importance of something
7. Short text of honour on a gravestone
8. A piece of writing that uses humour to bring out faults in a person, an institution, or some idea
9. A figure of speech in which one thing is described in terms of another

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CHAPTER

Chills and Thrills

Thriller is a genre of literature, film, and television programming that uses suspense, tension and excitement as its main elements.

A. Tick (✓) the correct options.

- Captain Arthur Hasting is a friend of :
 (a) Perry Mason ☐ (b) Hercule Poirot ☐
- Perry Mason's secretary is :
 (a) Della Street ☐ (b) Hilda ☐
- In which Ian Fleming novel does James Bond first appear?
 (a) Casino Royale ☐ (b) Moonraker ☐
- He is a dealer in curious and antique jewellery and owns a Shop in Nancy Drew's hometown.
 (a) Salty Weatherby ☐ (b) Sam Weatherby ☐



B. Identify these famous writers of thrillers and mysteries and their popular creations.

- He created the series of adventures about brothers-cum detectives titled The Hardy Boys.
- He is a defence attorney in the detective fiction authored by Erle Stanley Gardner. He was also a part of a popular radio series and was featured in several movies.
- The famous Bourne series was authored by which famous writer?
- Created by Sir Arthur Conan Doyle, this famous detective operated from his apartment. A Study in Scarlet is among the most famous stories in which he appears.



Activity

Read any one thriller book and discuss its story in the class.

The Three Apples, a tale in The Thousand and One Nights, is the earliest known murder mystery and suspense thriller.





Full Form and Short Form

A. Fill in the blanks in this quiz on the full forms and abbreviations of some well-known international organizations.

1. Environmental Protection Agency
2. Association of Southeast Asian Nations
3. Amnesty International
4. FEMA
5. ISO
6. World Copyright Treaty
7. Missile Technology Control Regime
8. UNDP
9. ICJ
10. DPKO

B. Tick (✓) the correct option.

1. FAO
 - (a) Food and Agriculture Organization ☐
 - (b) Fertilizer and Agriculture Organization ☐
2. GDP
 - (a) Gross Development Proportion ☐
 - (b) Gross Domestic Product ☐
3. MNC
 - (a) Multinational Committee ☐
 - (b) Multinational Corporation ☐



Project Work

What do CRY, UNICEF, and HELP AGE stand for? Talk about the work they do.



Guess The Word

From time to time, new words and phrases that comes to be widely used in our every day communication are included in the English dictionary.

Hidden in the grid on the next page are some words and phrases. Read the clues and guess them. Circle the words on the grid and fill them in the boxes beneath the clues. One has been done for you.

- One who uses technology with great ease

T e c h s a v v y

- A brief but refreshing sleep during the work day that provides energy to work longer and more efficiently

o e a

- A fit of anger by the driver of an automobile, especially towards another motorist or pedestrian often resulting in quarrel or violence

o d a e

- A motto or a principle that is supposed to guide one's action

a r

- A humorous phrase to describe the ordinary postal system, in comparison to electronic mail

n l a

- A blog that contains video material

l g

- A person who tends to worry habitually and often needlessly

o r w r

- A computer controlled by a hacker without the owner's knowledge, which is made to send large quantities of data to a website, making it inaccessible to others

Z m e

- An important feature of a tool or gadget ensuring that it can be used with ease

s r d

10. A feeling of having experienced something before, while experiencing it for the first time

e **v**

11. A reincarnation in human form, or a computer user's representation of the self

v **a**

12. Relating to or concerned with wholes or the complete system, rather than parts

H **i** **c**

13. A large movie theatre complex with several separate screens

l **p**

14. A person who stays at home and takes care of the house and family

o **m** **r**

P	H	A	V	A	T	A	R	I	I	K	L	M	N	M	M	F	T	T	D
Z	O	M	B	I	E	P	L	E	G	I	C	P	R	U	A	L	E	E	E
F	A	D	G	V	T	H	U	D	R	E	Q	F	S	L	N	O	C	W	J
U	E	E	N	R	H	O	M	E	M	A	K	E	R	T	T	W	H	K	A
W	I	N	D	W	P	R	B	R	U	C	T	H	R	I	R	E	S	I	V
G	L	B	O	S	E	T	H	N	L	T	E	O	A	P	A	R	A	N	U
C	W	O	R	R	Y	W	A	R	T	F	R	L	S	L	B	P	V	L	O
O	V	L	L	A	G	E	R	P	I	I	A	I	T	E	E	A	V	P	B
M	I	E	R	T	R	A	I	H	P	R	I	S	O	X	D	L	Y	E	I
O	C	H	I	L	G	N	V	A	L	E	N	T	U	N	G	G	K	C	A
U	D	R	O	A	D	R	A	G	E	B	B	I	L	V	L	O	G	L	P
S	A	T	L	U	P	I	T	I	X	R	A	C	L	E	M	C	W	L	K
E	P	S	A	M	K	N	R	U	N	I	S	N	A	I	L	M	A	I	L
P	O	W	E	R	N	A	P	M	T	G	A	D	E	A	M	S	H	A	Z
U	S	E	R	F	R	I	E	N	D	L	Y	R	L	L	A	D	E	M	I

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CHAPTER

Quiz Session

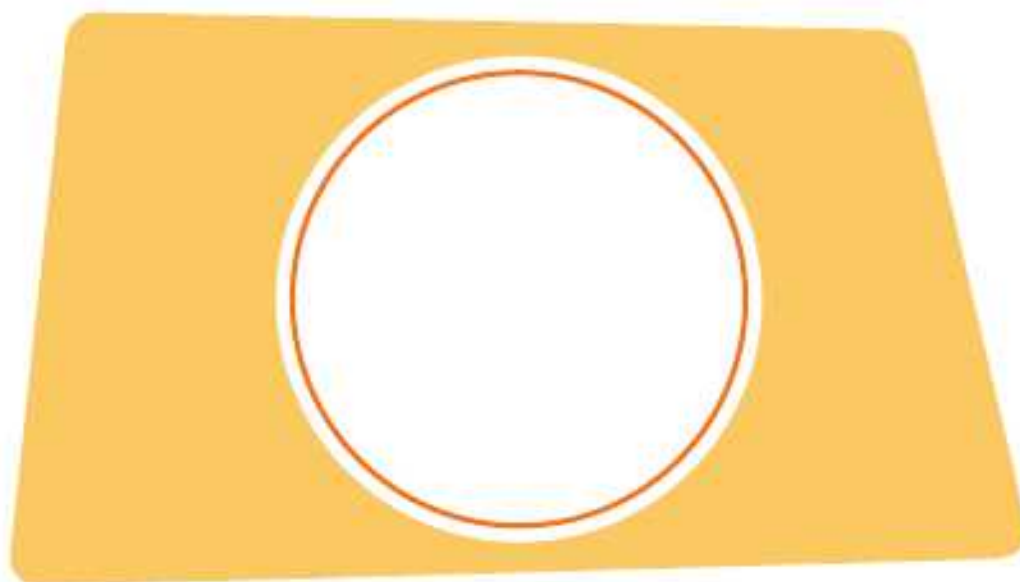
- Q.1.** Which collection of stories includes tales like 'Cinderella' and 'Little Red Riding Hood'?
- | | | | |
|-------------------------|--------------------------|--------------------|--------------------------|
| (a) Arabian Nights | ☐ | (b) Aesop's Fables | ☐ |
| (c) Grimm's Fairy Tales | ☐ | (d) Panchatantra | ☐ |
- Q.2.** In which book did the character "Mowgli" appear?
- | | | | |
|--------------------------------------|--------------------------|---------------------|--------------------------|
| (a) Alice's Adventures in Wonderland | ☐ | (b) The Jungle Book | ☐ |
| (c) The Wind in the Willows | ☐ | (d) Treasure Island | ☐ |
- Q.3.** Who wrote the famous play "Romeo and Juliet"?
- | | | | |
|-------------------------|--------------------------|-----------------|--------------------------|
| (a) William Shakespeare | ☐ | (b) Jane Austen | ☐ |
| (c) Charles Dickens | ☐ | (d) Leo Tolstoy | ☐ |
- Q.4.** What is the term used for a play that is intended to make the audience laugh?
- | | | | | | | | |
|-------------|--------------------------|------------|--------------------------|-----------|--------------------------|-------------|--------------------------|
| (a) Tragedy | ☐ | (b) Comedy | ☐ | (c) Drama | ☐ | (d) Mystery | ☐ |
|-------------|--------------------------|------------|--------------------------|-----------|--------------------------|-------------|--------------------------|
- Q.5.** What do you call a word that sounds the same as another word but has a different meaning and spelling?
- | | | | | | | | |
|-------------|--------------------------|-------------|--------------------------|-------------|--------------------------|---------------|--------------------------|
| (a) Synonym | ☐ | (b) Homonym | ☐ | (c) Antonym | ☐ | (d) Homophone | ☐ |
|-------------|--------------------------|-------------|--------------------------|-------------|--------------------------|---------------|--------------------------|
- Q.6.** Who wrote the famous book "To Kill a Mockingbird"?
- | | | | | | | | |
|--------------------|--------------------------|----------------|--------------------------|------------------|--------------------------|----------------|--------------------------|
| (a) J.R.R. Tolkien | ☐ | (b) Harper Lee | ☐ | (c) J.K. Rowling | ☐ | (d) Mark Twain | ☐ |
|--------------------|--------------------------|----------------|--------------------------|------------------|--------------------------|----------------|--------------------------|
- Q.7.** What does the expression 'spill the beans' mean?
- | | | | |
|---|--------------------------|------------------------------|--------------------------|
| (a) Taking out stolen beans from the pocket | ☐ | (b) Telling lies about beans | ☐ |
| (c) Telling the truth about beans | ☐ | (d) Telling a secret | ☐ |
- Q.8.** Which famous collection of stories includes the tales of 'Sindbad the Sailor' and 'Aladdin and the Magic Lamp'?
- | | | | |
|------------------|--------------------------|--------------------|--------------------------|
| (a) Jataka Tales | ☐ | (b) Arabian Nights | ☐ |
| (c) Panchatantra | ☐ | (d) None of these | ☐ |
- Q.9.** Our national song 'Vande Mataram' was first featured in a book called 'Anandamath'. Who wrote it?
- | | | | |
|-------------------------------|--------------------------|-------------------------|--------------------------|
| (a) Subhas Chandra Bose | ☐ | (b) Rabindranath Tagore | ☐ |
| (c) Bankim Chandra Chatterjee | ☐ | (d) Satyajit Ray | ☐ |
- Q.10.** In which book did the character "Harry Potter" attend Hogwarts School of Witchcraft and Wizardry?
- | | | | |
|---|--------------------------|-----------------------|--------------------------|
| (a) The Chronicles of Narnia | ☐ | (b) The Hobbit | ☐ |
| (c) Harry Potter and the Sorcerer's Stone | ☐ | (d) Lord of the Rings | ☐ |



Number Logic

Try solving the following fun match puzzles in the shapes given below.

1. Try dividing a circle into 11 pieces using just 4 lines. Draw it out below.



2. Place 10 plates along the edges of a rectangle so that each edge has the same number of plates.





Coding : Decoding

In these questions, the letters in a word are replaced by certain other letters according to a specific rule to form its code. The candidate is required to detect the coding pattern/rule and answer the questions accordingly.

- In a certain code, TEACHER is written as VGCEJGT. How is CHILDREN written in that code?
 (a) EJKNEGTP ☐ (b) EGKNFITP ☐
 (c) EJKNFGTO ☐ (d) EJKNFTGP ☐
- In a certain code language, RUSTICATE is written as QTTUIDBSD. How would STATISTIC be written in that code?
 (a) RSBUITUHB ☐ (b) RSBUITUHB ☐
 (c) RSBUIRSJD ☐ (d) TUBUITUMB ☐
- If ROAST is coded as PQYUR in a certain language, then how will SLOPPY be coded in that language?
 (a) MRNAQN ☐ (b) NRMNQA ☐
 (c) QNMRNA ☐ (d) RANNMQ ☐
- If HEALTH is written as GSKZDG, then how will NORTH be written in that code?
 (a) OPSUI ☐ (b) GSQNM ☐
 (c) FRPML ☐ (d) IUSPO ☐
- In a certain code, BREAKTHROUGH is written as EAOURBRGHKT. How is DISTRIBUTION written in that code?
 (a) TISTBUONDIRI ☐ (b) STTIBUONRIDI ☐
 (c) STTIBUDIONRI ☐ (d) RISTTIBUDION ☐
- In a certain code language, BEAT is written as YVZG, then what will be the code of MILD?
 (a) ONRW ☐ (b) NOWR ☐
 (c) ONWR ☐ (d) NROW ☐
- In a certain code, FAVOUR is written as EBUPTS. How is DANGER written in that code?
 (a) CBFFDS ☐ (b) CBMHDS ☐
 (c) EBFHDS ☐ (d) EBHHFS ☐

8. If SUMMER is coded as RUNNER, the code for WINTER will be :
- (a) SUITER ☐ (b) VIOUER ☐
 (c) WALKER ☐ (d) SUFFER ☐
9. In a certain code, BASIC is written as DDULE. How is LEADER written in that code?
- (a) NGCFG T ☐ (b) NHCGGU ☐
 (c) OGD FHT ☐ (d) OHDGHU ☐
10. In a certain language, SIGHT is written as FVTUG. How is REVEAL written in the same language?
- (a) YNRIRE ☐ (b) DQHQM X ☐
 (c) FSJSOZ ☐ (d) ERIRNY ☐
11. In a certain code, BELIEF is written as AFKKDI. How is SELDOM written in that code?
- (a) RDKCNL ☐ (b) RFKENM ☐
 (c) RFKFNP ☐ (d) TFKENP ☐
12. If TRUTH is coded as SUQSTVSUGI, then the code for FALSE will be :
- (a) EGZBKMRDE ☐ (b) EGZKMRTDF ☐
 (c) EGZBKMR TDF ☐ (d) FGZBKNRTDF ☐
13. If DELHI can be coded as CCIDD, how would you code BOMBAY?
- (a) AJMTVT ☐ (b) AMJXVS ☐
 (c) MJXVSU ☐ (d) WXYZAX ☐
14. If BOMBAY is written as MYMYMY, how will TAMIL NADU be written in that code?
- (a) TIAT IATIA ☐ (b) MNUMNUMNU ☐
 (c) IAT IAT IAT ☐ (d) ALDALDALD ☐
15. In a certain code, TELEPHONE is written as ENOHPELET. How is ALIGATOR written in that code?
- (a) ROTAGILA ☐ (b) ROTAGAIL ☐
 (c) ROTAGILE ☐ (d) ROTEGILA ☐

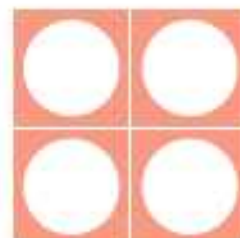


52

CHAPTER

Prickly Puzzles

1. The diameter of the circle is 4 cm. Calculate the area that is shaded.



2. Hidden in the grid are eight 7-letter words. Each word begins with the central **S**, and then you can keep on moving to the next letter in any direction. Each letter is used exactly once. What are the words?

G	N	L	K	R	T	E
E	I	E	O	X	A	E
K	R	N	I	H	M	N
E	A	Y	S	E	A	P
R	E	P	P	A	W	O
N	E	A	G	U	E	O
R	N	E	S	A	D	E

3. Using the letters

E	E	I	I	I	I	S	S	S	T	T	T
---	---	---	---	---	---	---	---	---	---	---	---

 complete this grid so that it reads the same across as down.

			R
		R	
	R		
R			

4. What comes next in this sequence?

A	A	A	B	D	E	E	E	E	F	F	F	G	H	H	H
I	J	K	K	K											



Number Series

Find the missing term in each of the following series.

1. 6, 15, _____, 45, 66, 91
(a) 25 (b) 26 (c) 27 (d) 28
2. 2, 5, 9, 19, 37, _____
(a) 73 (b) 75 (c) 76 (d) 78
3. 4, 8, 28, 80, 244, _____
(a) 278 (b) 428 (c) 628 (d) 728
4. 0, 6, 24, 60, 120, 210, _____
(a) 240 (b) 290 (c) 336 (d) 504
5. 1, 4, 27, 16, _____, 36, 343
(a) 25 (b) 87 (c) 120 (d) 125
6. 4, 6, 12, 14, 28, 30, _____
(a) 32 (b) 60 (c) 62 (d) 64
7. 1, 9, 25, 49, _____, 121
(a) 64 (b) 81 (c) 91 (d) 100
8. 4, 7, 12, 19, 28, _____
(a) 30 (b) 36 (c) 39 (d) 49
9. 11, 13, 17, 19, 23, 25, _____
(a) 26 (b) 27 (c) 29 (d) 37
10. 2, 5, 9, _____, 20, 27
(a) 14 (b) 16 (c) 18 (d) 24

11. 10, 18, 28, 40, 54, 70, ____

(a) 85



(b) 86



(c) 87



(d) 88



12. 120, 99, 80, 63, 48, ____

(a) 35



(b) 38



(c) 39



(d) 40



13. 22, 24, 28, ____, 52, 84

(a) 36



(b) 38



(c) 42



(d) 46



14. 0, 2, 8, 14, ____, 34

(a) 20



(b) 23



(c) 24



(d) 25



15. 1, 5, 13, 25, 41, ____

(a) 51



(b) 57



(c) 61



(d) 63



16. 2, 15, 41, 80, ____

(a) 111



(b) 120



(c) 121



(d) 132



17. 1, 4, 10, 22, 46, ____

(a) 64



(b) 86



(c) 94



(d) 122



18. 5, 6, 9, 15, ____, 40

(a) 21



(b) 25



(c) 27



(d) 33



19. 4, 12, 36, 108, ____

(a) 144



(b) 216



(c) 304



(d) 324



20. 1, 2, 3, 6, 9, 18, ____, 54

(a) 18



(b) 27



(c) 36



(d) 81



In each of the following questions, there is a certain relationship between two given words on one side, and one word is given on another side of :: while another word is to be found from the given alternatives, having the same relation with this word as the words of the given pair bear.

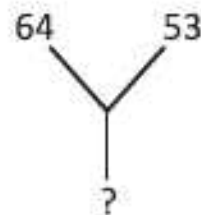
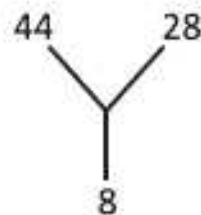
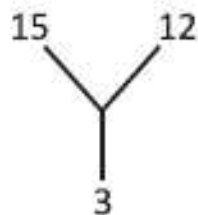
Choose the correct alternative.

1. Giant : Dwarf :: Genius : _____
 (a) Wicked ☐ (b) Gentle ☐
 (c) Idiot ☐ (d) Tiny ☐
2. Newspaper : Press :: Cloth : _____
 (a) Tailor ☐ (b) Textile ☐
 (c) Fibre ☐ (d) Mill ☐
3. Cattle : Herd :: Sheep : _____
 (a) Flock ☐ (b) Swarm ☐
 (c) Shoal ☐ (d) Mob ☐
4. Acting : Theatre :: Gambling : _____
 (a) Casino ☐ (b) Club ☐
 (c) Bar ☐ (d) Gym ☐
5. Mumbai : Maharashtra :: Trivandrum : _____
 (a) Kolkata ☐ (b) Gujarat ☐
 (c) Kerela ☐ (d) Sikkim ☐
6. Menu : Food :: Catalogue : _____
 (a) Rock ☐ (b) Newspaper ☐
 (c) Library ☐ (d) Books ☐
7. Pulp : Paper :: Hemp : _____
 (a) Basket ☐ (b) Yarn ☐
 (c) Rope ☐ (d) Cotton ☐
8. Dog : Bark :: Goat : _____
 (a) Bleat ☐ (b) Howl ☐
 (c) Grunt ☐ (d) Bray ☐

9. Food : Stomach :: Fuel : _____
 (a) Plane ☐ (b) Truck ☐
 (c) Engine ☐ (d) Automobile ☐
10. Moon : Satellite :: Earth : _____
 (a) Sun ☐ (b) Planet ☐
 (c) Solar System ☐ (d) Asteroid ☐
11. Cricket : Bat :: Hockey : _____
 (a) Field ☐ (b) Stick ☐
 (c) Player ☐ (d) Ball ☐
12. Bird : Wing :: Fish : _____
 (a) Gill ☐ (b) Fin ☐
 (c) Tail ☐ (d) Scale ☐
13. Coconut : Shell :: Letter : _____
 (a) Letter Box ☐ (b) Stamp ☐
 (c) Mail ☐ (d) Envelope ☐
14. Good : Bad :: Virture : _____
 (a) Blame ☐ (b) Sin ☐
 (c) Despair ☐ (d) Vice ☐
15. Melt : Liquid :: Freeze : _____
 (a) Ice ☐ (b) Condense ☐
 (c) Solid ☐ (d) Crystal ☐
16. Safe : Secure :: Protect : _____
 (a) Conserve ☐ (b) Sure ☐
 (c) Guard ☐ (d) Lock ☐
17. Girl : Beautiful :: Boy : _____
 (a) Smart ☐ (b) Heroic ☐
 (c) Courageous ☐ (d) Handsome ☐
18. Chair : Furniture :: Shoe : _____
 (a) Socks ☐ (b) Footwear ☐
 (c) Leather ☐ (d) Cobbler ☐

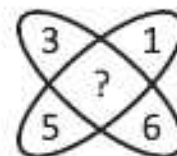
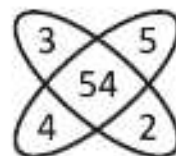
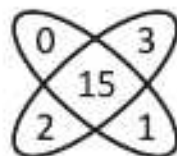
1. Find the missing number from the given responses in the following question.

- (a) 30
- (b) 13
- (c) 70
- (d) 117



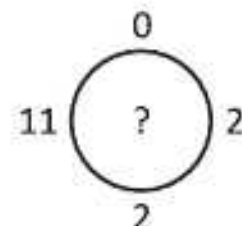
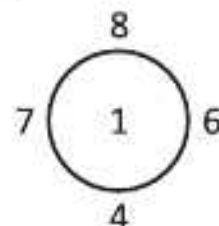
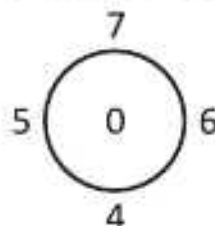
2. Find the missing term from amongst the options.

- (a) 35
- (b) 71
- (c) 72
- (d) 94



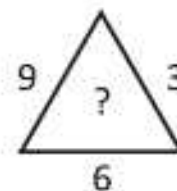
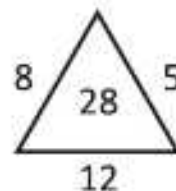
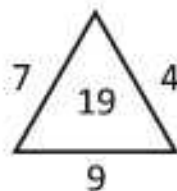
3. Select the missing number from amongst the options.

- (a) 8
- (b) 9
- (c) 10
- (d) 11



4. Choose the missing term to complete the given series.

- (a) 27
- (b) 21
- (c) 17
- (d) 28



5. Use the rule of the pattern and find the missing number which will replace the question mark.

- (a) 6
- (b) 9
- (c) 14
- (d) 15

4	9	2
3	5	7
8	1	?



Answer Sheet

Unit-1 : The Natural World

Chapter-1

1. Resurrection Fern, 2. Joshua Tree, 3. Elephant Foot Tree, 4. Mistletoe, 5. Banyan Trees, 6. Epiphytes, 7. Mangroves, 8. Arctic Poppy.

Chapter-2

1. Gangotri, 2. Old Faithful, 3. Grand Canyon, 4. Stalagmite, 5. Cotopaxi, 6. Atacama, 7. Stalactite, 8. Titicaca, 9. Bora Bora, 10. Giant's Causeway.

Chapter-3

1. Crab, 2. Lantern Fish, 3. Manatee, 4. Shark, 5. Jelly Fish, 6. Whale Shark, 7. Star Fish, 8. Puffer Fish, 9. Scallop, 10. Giant Clam.

Chapter-4

1. Sierra Nevada, 2. Khao Sok, 3. Sherwood Forest, 4. Deciduous Forest, 5. Forest Ecology, 6. Boreal Forest.

Chapter-5

1. Orangutan, 2. Ocelot, 3. Horns, 4. Siberian, 5. Numbat, 6. Chiru, 7. Ivory, 8. Dugong.

Chapter-6

1. a, 2. a, 3. b, 4. a.

Chapter-7

1. Arctic Fox, 2. Hippopotamus, 3. Burmesepython, 4. Frill Lizard, 5. Sloth, 6. Poison Dart Frog, 7. Badger, 8. Weddell Seal.

Chapter-8

(A) : 1. Southern, 2. Lambert Glacier, 3. Roald Amundsen, 4. Vinson Massif, 5. Nunataks.

(B) : 1. (✓), 2. (×), 3. (✓), 4. (✓), 5. (✓).

(C) : 1. c, 2. b, 3. b, 4. c, 5. c.

(D) : 1. Krill, 2. Baleen Whale, 3. Fur Seal, 4. South Polar Skua, 5. Emperor Penguin, 6. Sea Lion.

Chapter-9

1. Avalanche, 2. Hurricane, 3. Earthquake, 4. Chile, 5. Tornado, 6. Eye, 7. Saffir-Simpson, 8. Kosi, 9. Tohoku, 10. Katrina.

Chapter-10

1. CFC, 2. Antarctica, 3. (b) Methyl isocyanate, 4. (c) Both a and b, 5. The Cancun Agreement, 6. Step Farming, 7. (a) Malaria, 8. (c) Compact Fluorescent Lamp, 9. Methane.

Chapter-11

Across : 4. PRODUCERS, 5. FOODWEB, 8. HOTSPOTS, 9. CONSUMERS, 10. HERBIVORE, 12. DECOMPOSERS, 14. PHOTOSYNTHESIS, 16. SYMBIOSIS, 17. EVAPORATION, 18. ABIOTIC, 19. BIODIVERSITY, 20. PRECIPITATION.

Down : 1. ECOSYSTEM, 2. PREY, 3. SCAVENGERS, 6. CHLOROPHYLL, 7. CARNIVORE, 11. TRANSPIRATION, 13. HABITAT, 14. PREDATOR, 15. BIOTIC.

Unit-2 : The World of Science

Chapter-12

1. (i), 2. (iii), 3. (i), 4. (iv), 5. (i), 6. (i), 7. (ii), 8. (ii), 9. (i), 10. (ii)

Chapter-14

1. C.V. Raman, 2. Charles Babbage, 3. Benjamin Franklin, 4. Charles Darwin, 5. Issac Newton.

Chapter-15

A. 1. Ganymede, 2. Space Crafts, 3. Neptune, 4. Uranus, 5. Columbia, 6. Sun.

B. 1. Mercury, 2. Venus, 3. Mars, 4. Ozone, 5. Uranus, 6. Sputnik, 7. Meteorites, 8. Cosmology, 9. Sea of Tranquility, 10. Buran.

Chapter-16

1. b, 2. e, 3. g, 4. a, 5. f, 6. c, 7. i, 8. d, 9. h, 10. j

Chapter-17

1. Windows, 2. Silicon, 3. Shift, 4. Hacker, 5. Monitor, 6. UNIVAC, 7. Megabyte, 8. IBM, 9. Modem.

Chapter-18

Across : 1. ECZEMA, 3. ARTHRITIS, 4. MYOPIA, 5. DYSLEXIA

Down : 2. SCURVY, 6. HAYFEVER, 7. ASTHMA, 8. MIGRAINE, 9. TETANUS.

Chapter-19

1. Dermatologist, 2. Geriatrician, 3. Cardiologist,
4. Gastroenterologist, 5. Endocrinologist, 6. Haematologist,
7. Nephrologist, 8. Oncologist, 9. Paediatrician,
10. Pulmonologist.

Unit-3 : Our Country : India

Chapter-22

1. Salim Ali, 2. Dr. J.J. Chinoy, 3. Dr. C.V. Raa,
4. Meghaad Shah, 5. Dr. A.J. Abdul Kalam, 6. Dr. Vikram
- Sarabhai, 7. Prof. R.R. Ramanathan, 8. Prof. U.R. Rao,
9. Dr. Jagdish Chandra Bose

Unit-4 : Sports and Entertainment

Chapter-24

1. b, 2. a, 3. b, 4. a, 5. b, 6. a.

Chapter-25

1. Wilma Rudolph, 2. Temba Tsheri Sherpa, 3. Arunima
- Sinha, 4. Jim Abbott.

Chapter-26

1. Chris Everts Mills, 2. Mark Spitz, 3. Ramanathan
- Krishnan, 4. Sugar Ray Robinson, 5. Geoffrey Boycott.

Chapter-27

1. Judo, 2. Sambo, 3. Kick Boxing, 4. Capoeira, 5. Tae
- Kwon Do.

Chapter-28

1. c, 2. a, 3. b, 4. a, 5. a, 6. a, 7. a.

Chapter-29

- A. 1. Paragliding, 2. Ice-climbing, 3. Sand boarding,
4. Surfing, 5. Bungee Jumping, 6. Skateboarding,
7. Kayaking, 8. Skiing.

- B. 1. a, 2. a, 3. b, 4. a, 5. a, 6. a

Chapter-30

1. George Clooney, 2. Shabana Azmi, 3. Shakira, 4. Bill
- Gates, 5. Jackie Chan.

Chapter-31

- (A) 1. c, 2. e, 3. a, 4. f, 5. b, 6. d.
- (B) Do yourself

Chapter-32

1. Ventriloquist, 2. Pirouette, 3. Aristotle, 4. Libretto,

5. Adance for two people, 6. India, 7. William
- Shakespeare.

Chapter-33

1. UP, 2. Chota Chetan, 3. Anakin Skywalker, 4. River
- Amazon, 5. Sound of Music, 6. Marilyn Monroe,
7. Gurinder Chadha, 8. Chachi 420, 9. Amitabh
- Bachchan, 10. China, 11. Ian Mckellen, 12. Mushu.

Chapter-34

1. c, 2. c, 3. b, 4. a, 5. b, 6. c, 7. a, 8. c.

Chapter-35

1. Ravi Shanker, 2. I. Subramaniam, 3. Bhupen
- Hazarika, 4. Louis Banks, 5. Sachin Dev Burman.

Unit-5 : Around the World

Chapter-36

1. (i), 2. (ii), 3. (i), 4. (iv), 5. (iv), 6. (iii), 7. (iv), 8. (iii), 9. (i),
10. (ii)

Chapter-37

1. Mark Zuckerberg, 2. Reserve Bank of India, 3. Dalal
- Street, 4. Lehman Brothers, 5. Larry Page and Sergey
- Brin, 6. Amul.

Chapter-38

1. Egypt, 2. Cha Cha Cha, 3. India, 4. Glastonbury,
5. Paris, 6. Oktoberfest, 7. Joan of Arc, 8. Ramlila,
9. Indonesia, 10. Mali.

Chapter-39

1. b, 2. g, 3. l, 4. e, 5. c, 6. d, 7. a, 8. f, 9. j, 10. k, 11. i,
12. h.

Chapter-40

1. Jatiyo Smriti Sudho, 2. India Gate, 3. Monument to
- Peter I, 4. Vivekanand Rock Memorial, 5. Terracota
- Army, 6. Tomb of the Unknown Soldier.

Chapter-41

1. Sinhala, 2. Austria, 3. Japan, 4. Francis Scott Key,
5. United Kingdom.

Chapter-42

1. Calvin Klein, 2. British Airlines, 3. BMW, 4. DHL,
5. Hewlett Packard, 6. Cisco, 7. Nike, 8. FedEx, 9. Apple,
10. Gillette, 11. Sony, 12. Nokia, 13. Nissan, 14. Fiat,
15. Ford, 16. LG, 17. Walmart, 18. Philips.

Chapter-43

(A) : 1. e, 2. d, 3. a, 4. f, 5. b, 6. g, 7. c.

(B) : 1. c, 2. b, 3. b, 4. a.

Chapter-44

1. Junoon, 2. Severe Acute Respiratory Syndrome (SARS), 3. Israel, 4. India, 5. Iraq, 6. Atlantis, 7. Dennis Tito.

Unit-6 : Language and Literature

Chapter-45

1. Prologue, 2. Oxymoron, 3. Alliteration, 4. Syntax, 5. Allegory, 6. Hyperbole, 7. Epitaph, 8. Satire, 9. Simile.

Chapter-46

(A) : 1. b, 2. a, 3. a, 4. b.

(B) : 1. Edward Stratemeyer, 2. Perry Mason, 3. Robert Ludlum, 4. Sherlock Holmes.

Chapter-47

(A) : 1. EPA, 2. ASEAN, 3. AI, 4. Federal Emergency Management Agency, 5. International Organization for Standardization, 6. WCT, 7. MTCR, 8. United Nations Development Programme, 9. International Court of Justice, 10. Department of Peace Keeping Operations.

(B) : 1. a, 2. b, 3. b.

Chapter-48

1. Tech Savvy, 2. Power Nap, 3. Road Rage, 4. Mantra, 5. Snail Mail, 6. Vlog, 7. Worrywart, 8. Zombie, 9. User-friendly, 10. Deja vu, 11. Avatar, 12. Holistic, 13. Multiplex, 14. Homemaker.

Chapter-49

1. (c), 2. (b), 3. (a), 4. (b), 5. (d), 6. (b), 7. (d), 8. (b), 9. (c), 10. (c)

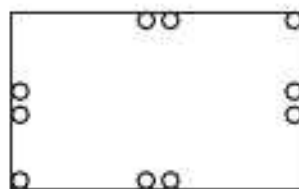
Unit-7 : Brain Teaser

Chapter-50

1.



2.



Chapter-51

1. d, 2. b, 3. c, 4. b, 5. c, 6. d, 7. b, 8. b, 9. b, 10. d, 11. c, 12. c, 13. d, 14. b, 15. a.

Chapter-52

1. 13.76 sq. cm, 2. Sixteen; Shampoo; Seaweed; Sausage; Spanner; Speaker; Syringe; Shorkel,

3.

S	t	i	R
T	i	R	e
I	R	i	s
R	e	s	t

4.

L	L
---	---

Chapter-53

1. d, 2. b, 3. d, 4. c, 5. d, 6. b, 7. b, 8. c, 9. c, 10. a, 11. d, 12. a, 13. a, 14. c, 15. c, 16. d, 17. c, 18. b, 19. d, 20. b.

Chapter-54

1. c, 2. d, 3. a, 4. a, 5. c, 6. d, 7. c, 8. a, 9. c, 10. b, 11. b, 12. b, 13. d, 14. d, 15. c, 16. c, 17. d, 18. b.

Chapter-55

1. (b), 2. (c), 3. (d), 4. (b), 5. (a)